

A minimal chemo-mechanical Markov model for rotary catalysis of F₁-ATPase

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F₁-ATPase, the catalytic domain of ATP synthase, is pivotal for mechano-chemical energy conversion in mitochondria. Aiming at a minimal yet quantitative and thermodynamically consistent model for its rotary catalysis mechanism, here we developed a chemo-mechanical Markov model incorporating essential conformational and chemical degrees of freedom. By systematically evaluating over 14,000 model variants via Bayesian inference and cross-validation, we find that a fully functional minimal model requires four functionally distinct β -subunit conformations. Our model reconciles the decade-long bi-site versus tri-site controversy, showing that both pathways contribute depending on ATP concentration. Furthermore, our model suggests a Brownian-ratchet-like mechanism that explains the observation that one ATP hydrolysis event can trigger larger than 120° rotations, thereby explaining seemingly over 100% efficiency. Beyond this prototypic example of a complex biomolecular machine, our approach should enable one to study other enzymatic mechanisms that implement close coupling between conformational motions, substrate binding, and chemical reactions.

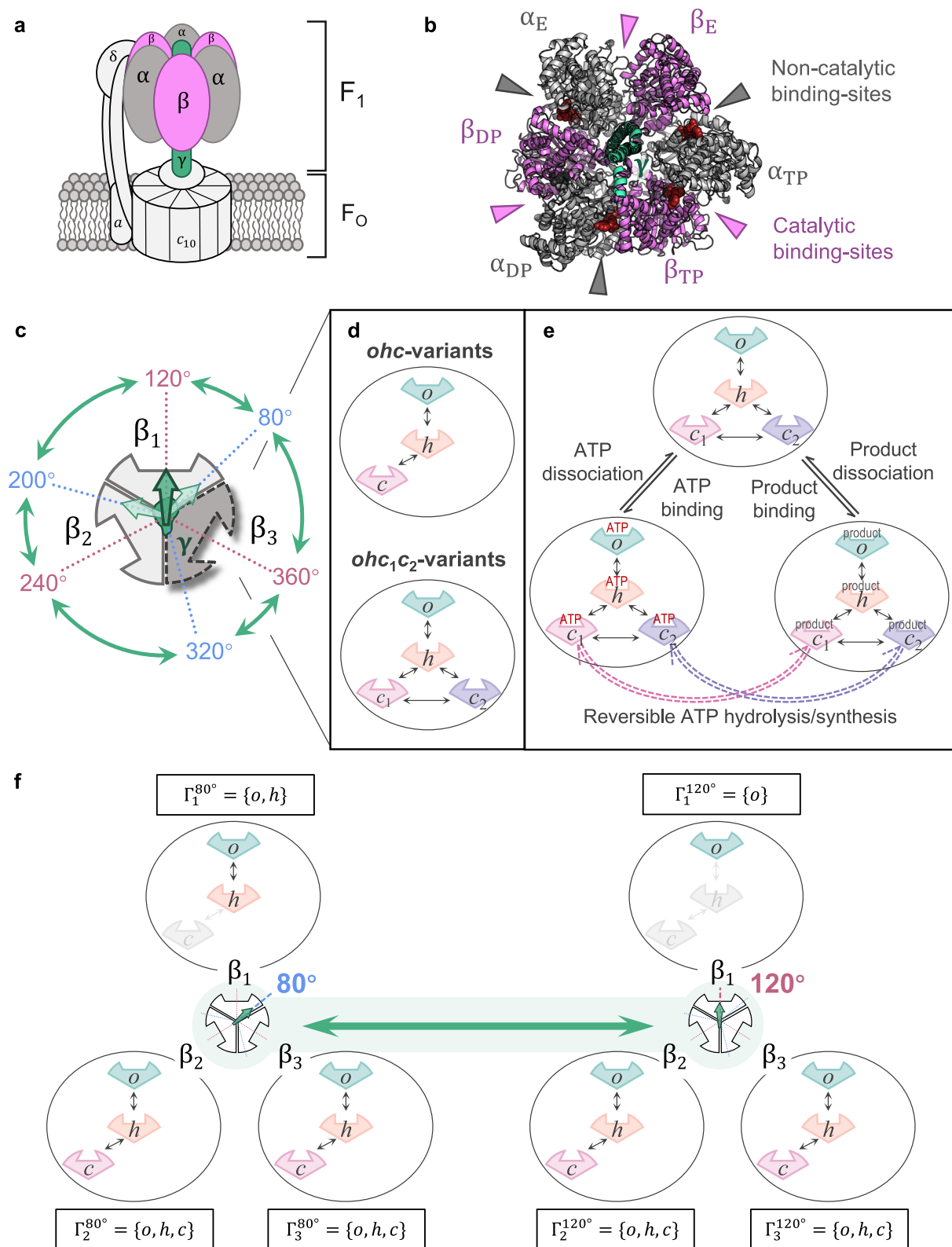
The F₁F₀-ATP synthase (ATP synthase) is a highly conserved biological rotary motor that synthesizes ATP from ADP and inorganic phosphate (Pi), using energy derived from the proton concentration gradient across membranes^{1–3}. The enzyme comprises two rotary motors, the membrane-embedded F₀ motor and the soluble F₁-ATPase, connected by a rotating central stalk (Fig. 1a). In vivo, ATP synthase typically operates in the synthesis mode. Here, proton flow through the F₀ motor drives the clockwise rotation of the central stalk, which in turn drives ATP synthesis within the three catalytic sites of the F₁-ATPase domain. Here we focus on the minimal $\alpha_3\beta_3\gamma$ structure of F₁-ATPase (Fig. 1b). This core structure consists of a stator ring formed by three α - and three β -subunits (gray and purple, respectively), with the γ -subunit (green) located centrally as an axle, forming part of the central stalk^{4,5}. Each β -subunit forms a catalytic site at an α - β interface. Remarkably, F₁-ATPase can reach ~100% efficiency^{6–10}. Accordingly, the F₁-ATPase can also operate in reverse (the hydrolysis) mode, where ATP hydrolysis within the catalytic sites drives rotation of the γ -subunit^{8,11–15}.

To rationalize this rotary catalysis, Paul Boyer proposed the initial binding change model^{16–18}, in which rotation of the γ -subunit

induces alternating conformational changes within the three β -subunits. These different conformations exhibit different binding affinities for ATP and ADP^{19–22}, thereby driving major catalytic steps such as substrate ATP binding, bound ATP hydrolysis, and product ADP/Pi dissociation. The tight coupling between these conformational and chemical degrees of freedom (DOFs) is at the core of this model. Recognized as a universal principle underlying rotary catalysis, this tight coupling mechanism was later generalized to the closely related V- and A-ATPases, grounded in the evolutionary, structural, and functional homology between the F-ATPase subunits (α , β , γ) and their V/A-ATPase counterparts (B, A, DF)^{23–30}.

Over time, the original binding change model¹⁷ has been repeatedly modified to accommodate further experimental observations, leading to a family of related models^{5,8,20,22,31–35}. However, despite decades of intensive research and a wealth of experimental data, there is yet no consensus on the precise mechanism of F₁-ATPase. Particularly, several questions remain unresolved, reflected in the differences among these models: (i) how many functionally distinct conformational states of the β -subunits are required for F₁-ATPase function

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(three^{16,20}, four^{22,31,32}, or potentially more^{5,36}); (ii) how many catalytic sites are simultaneously active during catalysis (bi-site^{6,8,16,37–39} versus tri-site^{15,20–22,32,34,40} mechanisms); (iii) how are the static structural snapshots from cryoEM and X-ray crystallography correlated with the functional dwells in F₁-ATPase catalytic cycle.

Methodologically, these earlier models typically focused on very few pre-assumed chemo-mechanical states and catalytic pathways, often motivated by biochemical intuition or static structures. This

focus has made it challenging to achieve a fully quantitative and thermodynamically consistent description of the rather complex F₁-ATPase system. Complementing these models, highly coarse-grained continuum ratchet models^{41–43} have been analyzed, and coarse-grained³⁶, atomistic^{44–49}, and QM/MM simulations^{50,51} have been performed. While these studies have revealed valuable mechanistic insights into rotary catalysis, they have not yet provided a complete thermodynamic description of the full catalytic cycle. Ultimately, a

Fig. 1 | The chemo-mechanical Markov model of F₁-ATPase. **a** Schematic of the F₁F₀-ATP synthase, showing the membrane-embedded F₀ sector and the soluble F₁ catalytic sector (F₁-ATPase). **b** Top-down view of the $\alpha_3\beta_3\gamma$ structure of F₁-ATPase (PDB: 1BMF), highlighting the three catalytic sites (purple arrows) and three non-catalytic sites (gray arrows) at the α - β interfaces. **c–e** Illustration of the DOFs defining the Markov states (Table 1). **c** The orientation of the central γ -subunit is discretized into six states corresponding to the catalytic (e.g., 80°) and ATP-waiting (e.g., 120°) dwells observed in single-molecule experiments. The γ -subunit rotates in discrete substeps between every two adjacent orientations. **d** Each β -subunit converts between three ($\{o, h, c\}$ for *ohc*-variants) or four ($\{o, h, c_1, c_2\}$ for *ohc₁c₂*-variants) distinct conformations. Direct transitions between open (*o*) and closed (*c*, *c₁*, *c₂*) conformations are disallowed, positioning the half-closed (*h*)

conformation as a mandatory intermediate. **e** Each β -subunit converts between three binding states: apo (E), ATP-bound (T) and product-bound (D). Reversible ATP hydrolysis/synthesis (dashed arrows) occurs only when the β -subunit is in a catalytically active conformation. **f** Visualization of the γ - β restrictions (the hyperparameter $\{\Gamma_k^\phi\}$) that define a specific model variant. This visualized example corresponds to the *ohc-w* variant (Table 2). The set of accessible conformations (e.g., $\{o\}$, $\{o, h\}$, or $\{o, h, c\}$) is illustrated for each β -subunit at 80° (left) and 120° (right). In this example, at 120°, the γ -subunit sterically forces β_1 into the *o* conformation, while β_2 and β_3 are unrestricted. These restrictions demonstrate how the asymmetric γ -subunit results in different conformational ensembles of the three chemically identical β -subunits.

coherent framework integrating the transition rates with the free energies for all possible combinations of the DOFs has not been established.

More fundamentally, the classical binding change models typically provide a macroscopic, ensemble-average view of rotary catalysis, and do not explicitly incorporate the inherent stochasticity at the molecular level. This perspective tends to intuitively align with a power-stroke mechanism for the chemo-mechanical energy transduction, assuming ATP hydrolysis directly generates the torque needed for unidirectional rotation. Nevertheless, extensive single-molecule experiments demonstrate large rotational fluctuations of the central stalk, pointing to a highly stochastic energy transduction process^{8,13–15,23,52,53}. Accordingly, a Brownian ratchet mechanism has been proposed: instead of directly exerting torque on the central stalk, ATP hydrolysis induces affinity changes in the catalytic subunits that directionally rectify these random thermal fluctuations^{29,30}. How to quantitatively reconcile the classical binding change paradigm with the stochastic nature of these microscopic molecular machines remains an open question^{8,15,29,30,52}.

In this work, we construct a minimal chemo-mechanical Markov model that unifies the conformational and chemical DOFs governing F₁-ATPase rotary catalysis to quantitatively capture its inherent stochasticity. To simultaneously explain a heterogeneous set of experimental data—including measured titration curves²¹, near 100% chemo-mechanical coupling efficiency⁸, and the consensus chemo-mechanical coupling scheme^{8,15,33,54–57}—we systematically evaluated various model variants via extensive Bayesian parameter inference and cross-validation. This systematic model comparison reveals that a fully functional model requires at least four, rather than three, distinct β -subunit conformations, contrary to previous assumptions^{16,20}. Furthermore, we find that a nucleotide-bound β -subunit energetically favors the closed conformation, thus helping to resolve a previous controversy^{46,48,58}. Remarkably, our model also reconciles the bi-site versus tri-site controversy by showing that both pathways contribute depending on ATP concentration. By construction, our model provides a quantitative understanding of the thermodynamics, kinetics, and inherent stochasticity of F₁-ATPase function.

Results

Construction of a chemo-mechanical Markov model

To establish our chemo-mechanical Markov model, we specifically focus on the well-characterized F₁-ATPases from *E. coli* and *Bacillus* PS3 (EF₁, TF₁), because the mechanistic details of F₁-ATPases from other species may differ, e.g., in their sub-stepping behaviors^{8,15,59–63}. Hereafter, unless otherwise specified, F₁-ATPase refers to EF₁ and TF₁. Below, we outline the conceptual framework of this model, while the rigorous mathematical formulations are provided in Methods.

In this model, a distinct chemo-mechanical state of F₁-ATPase (a Markov state) is specified by the orientation of the γ -subunit, and the conformational and binding states of the three β -subunits (Eq. (1)), totaling seven DOFs listed in Table 1.

First, we consider the rotation of the γ -subunit. For EF₁ and TF₁, this rotation is known to proceed via well-characterized 80° and 40° substeps^{8,15,54,59}, resulting in the six different γ -subunit orientations in our model (Fig. 1c).

Second, we consider for each β -subunit transitions between several conformational states (Fig. 1d). Crystal and cryoEM structures of F₁-ATPase reveal three clusters of clearly distinct conformations, often termed open (β_E), half-open/half-closed (β_{HO}/β_{HC}), and closed (β_{TP}/β_{DP})^{4,5,32,64,65}. Accordingly, we include three distinct conformations in our model, denoted as *o*, *h*, and *c* respectively. While tentatively associated with the experimentally resolved structures, these conformations are defined in our model by their functional roles: (1) the *o* conformation embeds a widely open catalytic site of very low nucleotide binding affinity that awaits substrate ATP binding; (2) the *c* conformation is the catalytically active state where ATP hydrolysis and synthesis occur reversibly and almost in equilibrium ($K_{eq} \approx 1^{6,66–68}$); (3) the *h* conformation is an intermediate state between the *o* and *c* conformations.

It is proposed that the two closed conformations, β_{TP} and β_{DP} , despite high structural similarity, are also functionally distinct, due to fine details at the catalytic interface between adjacent α - and β -subunits, e.g., involving the “arginine finger”^{4,5,32,67,69,70}. To systematically address the minimal number of functionally distinct conformations required for F₁-ATPase function, we tested two hypotheses—three-conformation versus four-conformation—by constructing two major classes of model variants (Fig. 1d): the *ohc*- and *ohc₁c₂*-variants, assuming one (*c*) or two functionally distinct closed conformations (*c₁*, *c₂*). Further, for the *ohc₁c₂*-variants, because it is controversially debated whether or not both *c₁* and *c₂* are catalytically active^{5,71,72}, we tested both hypotheses by investigating *ohc₁c₂*-variants and *ohc₁c₂*-variants, where the asterisks (*) denote the catalytically active conformations.

Third, we consider for each β -subunit conversion between three binding states (Fig. 1e)^{14,5,32,64,65}: an apo state (E), an ATP-bound state (T), and a single product-bound state (D) that merges the ADP-, Pi-, and ADP+Pi-bound states. This merged D state, a simplification also adopted in a recent theoretical analysis⁷³, results in an effective description of the kinetics of two chemical dissociation events, namely the dissociation of ADP and of Pi (further explained below).

As a key conceptual step, we a priori include all combinatorially possible combinations of the seven DOFs, resulting in $6 \times (3 \times 3)^3 / 3 = 1458$ (*ohc*-variants) or $6 \times (4 \times 3)^3 / 3 = 3456$ (*ohc₁c₂*-variants) Markov states in our model; here, the counts already take into account the three-fold rotational symmetry of F₁-ATPase, hence the division by three. Further, we restrict the transitions between these 1458 (or 3456) Markov states to those involving the change of only one DOF at a time, and, crucially, assume all these transitions to be reversible. These elementary transitions, as illustrated in Fig. 1c–e, include stepwise rotations of the γ -subunit in both directions, conformational transitions of each individual β -subunit between the three or four conformations, binding and dissociation of ATP or product, and reversible ATP hydrolysis/synthesis in a catalytically active β -subunit.

Table 1 | Degrees of freedom (DOFs) of our chemo-mechanical Markov model

DOFs	Possible values	Experimental justification
Orientation of the γ -subunit (ϕ)	$\phi = 80^\circ, 200^\circ, 320^\circ$, (catalytic dwells) $120^\circ, 240^\circ, 360^\circ$ (ATP-waiting dwells)	Single-molecule experiments ^{8,15,59}
Conformational states of the β -subunits ($C^{(k)}$, $k = 1, 2, 3$)	$C^{(k)} \in \{o, h, c\}$ (<i>ohc</i> -variants) or $C^{(k)} \in \{o, h, c_1, c_2\}$ (<i>ohc</i> ₁ <i>c</i> ₂ -variants)	Observation of several β -subunit conformations in crystal and cryoEM structures ($\beta_E, \beta_{TP}/\beta_{DP}, \beta_{HC}/\beta_{HO}$) ^{4,5,32,64,65}
Binding states of the β -subunits ($B^{(k)}$, $k = 1, 2, 3$)	$B^{(k)} = E$ (empty), T (ATP-bound), D (product-bound)	$\beta_E, \beta_{TP}/\beta_{DP}, \beta_{HC}/\beta_{HO}$ observed in different binding states ^{4,5,32,64,65}

The table lists the possible values of the seven DOFs that specify a Markov state (Eq. (1)) and their experimental justification.

Notably, our definition of elementary transitions inherently decouples conformational changes from chemical events like binding/dissociation or hydrolysis/synthesis. Without losing generality, such formal decoupling facilitates easier modeling of complex, multi-step catalytic processes as sequences of elementary events. For example, we describe product release in our model as a sequence of conformational changes (opening of the catalytic site) and subsequent chemical dissociation of the products. The fact that the former must precede the latter is described in our model by a very low product dissociation rate from a closed catalytic site (Supplementary Equation (5) and Supplementary Table 1). In principle, ADP- and Pi-release could be described separately and similarly; however, aiming at a minimal yet kinetically sufficiently accurate model, we decided to merge the ADP-, Pi-, and ADP+Pi-bound states into a single product-bound state D⁷³. In this approximation, the rates for dissociation of the two products are combined into an effective rate, governed by the slower of these two individual rates. This approximation is supported by experiments suggesting that only one of these two dissociation events is indeed rate-limiting^{15,55–57,74}.

By including all possible Markov states and reversible elementary transitions (Fig. 1c–e), our model explores all potential catalytic pathways without a priori bias, in contrast to previous binding change models^{5,8,16,20,22,31–35}. To quantitatively describe the dynamics of this Markov model (Eq. (2)), a transition rate needs to be specified for each elementary transition (Supplementary Table 2). Following transition state theory (Eq. (5)), the rate coefficient is determined by the free energy barrier between the two connected states; further, as required by the principle of local detailed balance, the rate coefficient of the backward transition is related to the forward rate coefficient by the difference between the free energies of the two states (Eq. (6)). However, to compare our model against experimental data characterizing steady-state properties of F₁-ATPase, we evaluated relevant model predictions under non-equilibrium steady-state (NESS) conditions (Methods), mimicking typical experimental setups where F₁-ATPase operates in contact with large external reservoirs maintaining constant ATP and ADP concentrations^{8,19,21}. In NESS, while local detailed balance is preserved for every individual elementary transition, the sustained chemical potential difference between ATP and ADP (Eq. (7)) breaks the global detailed balance of the system, driving a net flux through the network of states that manifests as observable ATP turnover and rotation.

We next decomposed the state free energies that dictate the transition rates into several contributions (Eq. (2)). Conformational free energies and binding free energies of individual β -subunits are included, characterizing the thermodynamic properties of the several functionally distinct conformations. Note that these energy terms also implicitly account for local interactions between adjacent α - and β -subunits at the catalytic interface^{4,5,32,67,69,70}. Because single-molecule and structural studies^{1,2,4,9,70,75} clearly point to a dominant role of the direct interaction between the γ - and β -subunits in governing F₁-ATPase rotary catalysis, we decided to include these γ - β interactions,

but not the long-range allosteric interactions across the $\alpha_3\beta_3$ stator ring. Whereas these allosteric interactions are proposed to facilitate residual activity of γ -depleted and γ -truncated constructs of F₁-ATPase^{76–79}, they are much weaker than the γ - β interactions, as evidenced by the greatly reduced turnover rates and coupling efficiency of the γ -depleted/truncated constructs^{76,77}. Along similar lines, and again aiming at a minimal model, we also refrained from including separate energy terms for local α - β interactions.

For defining the γ - β interaction energies (Eq. (4)), we note that structural^{4,5,32,64,65} and theoretical studies^{36,48} have suggested steric repulsions between the γ - and β -subunits: the γ -subunit pushes the N-terminal domain of the β -subunit that faces its bulge side outwards, forcing this β -subunit to adopt an open or partially open conformation. Accordingly, we defined an infinitely high interaction energy for some of the non-open conformation(s) of a β -subunit at a particular γ -orientation, which precludes these conformations from being populated. However, precisely which combinations of γ -orientations and conformations should be subjected to such restrictions is not immediately obvious from previous studies. Therefore, we introduced γ - β restrictions—an ensemble of accessible conformations defined for each β -subunit at each γ -orientation—as a key hyperparameter defining specific model variants (graphically illustrated in Fig. 1f, and formally defined in Methods). Even with symmetry considerations and plausible constraints, this hyperparameter leads to a large number of possibilities, e.g., more than 600 *ohc*-variants (Methods). To resolve this ambiguity, we first investigated several plausible *ohc*-variants, followed by a more systematic model exclusion analysis, as detailed below.

The above definitions of transition rates and state free energies have introduced ~20 parameters into our model, including conformational free energies and binding free energies assigned to each β -subunit conformation, as well as several free energy barriers (Methods).

No *ohc*-variant explains all available experimental data

To rigorously evaluate competing mechanistic hypotheses, we first tested whether a minimal model including only three functionally distinct β -subunit conformations *o*, *h*, and *c* (three-conformation hypothesis) could suffice to explain the available experimental data. Specifically, we examined three plausible *ohc*-variants (Table 2), asking if any of them, with optimized parameters, could simultaneously reproduce three key macroscopic properties defining F₁-ATPase rotary-catalysis function (training data, see Methods): (1) the turnover rates k_{cat} ¹⁹, (2) 100% chemo-mechanical coupling efficiency^{6,8,10,54}, and (3) average catalytic site occupancies under varying ATP/ADP concentrations (nucleotide titration curves)²¹. We employed a Bayesian approach (Methods) to broadly sample the parameter space and obtain a large ensemble of parameter sets covering potentially multimodal high-posterior regions.

Only the *ohc*-s variant yielded ~9000 parameter sets of similarly high posterior probabilities that successfully reproduced the training

Table 2 | Definition and evaluation of three plausible ohc-variants

Variant	γ - β restrictions (Γ_k^Φ)				Number of states	Training	Cross-validation test 1	Cross-validation test 2
ohc-w	ϕ	β_1	β_2	β_3	729	Failed	–	–
	80°	<i>o, h</i>	<i>o, h, c</i>	<i>o, h, c</i>				
	120°	<i>o</i>	<i>o, h, c</i>	<i>o, h, c</i>				
ohc-m	ϕ	β_1	β_2	β_3	648	Failed	–	–
	80°	<i>o, h</i>	<i>o, h, c</i>	<i>o, h, c</i>				
	120°	<i>o</i>	<i>o, h</i>	<i>o, h, c</i>				
ohc-s	ϕ	β_1	β_2	β_3	243	Succeeded	Succeeded	Failed
	80°	<i>o</i>	<i>o, h</i>	<i>o, h, c</i>				
	120°	<i>o</i>	<i>o</i>	<i>o, h, c</i>				

For each variant, the table lists its defining γ - β restrictions, the number of resulting Markov states, and the outcome of the training and cross-validation procedures (Methods). Training: A Bayesian approach was used to find parameter sets that reproduce the training data. Cross-validation: The trained parameter sets were tested against two sets of validation data, first (cross-validation test 1) the observed population shift between dwells⁸ and second (cross-validation test 2) the consensus chemo-mechanical coupling scheme (Table 6 and Supplementary Fig. 2). “Succeeded” or “Failed” indicates whether parameter sets consistent with the respective data were found.

data within experimental uncertainty (Fig. 2a, b). Notably, some predicted chemo-mechanical coupling efficiencies exceed 100% (Fig. 2a). This does not violate energy conservation, because this efficiency is defined as a ratio of rates (Eq. (12)) rather than energies, consistent with previous experimental observations⁸ (mechanistic explanation provided in Discussion).

To quantitatively assess agreement with experimental ADP titration curves²¹, we derived apparent ADP binding affinities by fitting the curves (Fig. 2b) to an equilibrium binding model (Eq. (17), see Methods). For the *ohc-s* variant (Fig. 2c), the predicted apparent affinities (pink, orange and green curves) agree well with the experimental values (horizontal lines). In contrast, the *ohc-w* and *ohc-m* variants disagree with the experimental values (Supplementary Fig. 1), and were therefore excluded from further analysis.

Training the *ohc-s* variant provided two key insights. First, despite reproducing the training data equally well, the ~9000 trained parameter sets predicted several distinct mechanistic scenarios, highlighting the need for additional experimental data to constrain our model. Second, and remarkably, across all scenarios, the catalytically active, closed conformation *c* is the state of lowest free energy for a nucleotide-bound β -subunit (Fig. 2f). Thus, unless sterically restricted by the γ -subunit, a nucleotide-bound β -subunit always remains closed. This energetic property is likely a necessary condition for our model to achieve near 100% efficiencies (Discussion).

To identify which mechanistic scenario is consistent with experimental observations, we cross-validated the model against experimental data that characterize mechanistic details of F_1 -ATPase at a more microscopic level (validation data, see Methods).

The first cross-validation test examined the population shift between the catalytic (γ -orientation ~80°) and ATP-waiting dwells (~120°). Experimentally, the dominant state shifts from ATP-waiting to catalytic dwells as ATP concentration ([ATP]) increases⁸. The trained parameter sets split into four groups predicting different trends (Fig. 2d). Only the first group correctly showed 120°-states and 80°-states dominating at low and high [ATP], respectively (leftmost panel); the other three groups were therefore discarded.

The second cross-validation test evaluated whether the 750 parameter sets selected from the first group (darker pink and blue curves) reproduce the γ -orientations of ATP binding, hydrolysis, and product dissociation steps, as defined by a consensus chemo-mechanical coupling scheme (see Methods and Supplementary Fig. 2; synthesized from a broad range of experimental studies^{8,15,20–22,32–34,54–57,65–67,70,75} as reviewed in Supplementary Note 2). While this consensus scheme adopts a deterministic view where each catalytic step is assigned to a specific γ -orientation, our model inherently predicts stochastic probability distributions for these steps to

occur at different γ -orientations (Methods, see particularly Eq. (13)). Therefore, we tested whether these probabilities concentrated at the consensus γ -orientations.

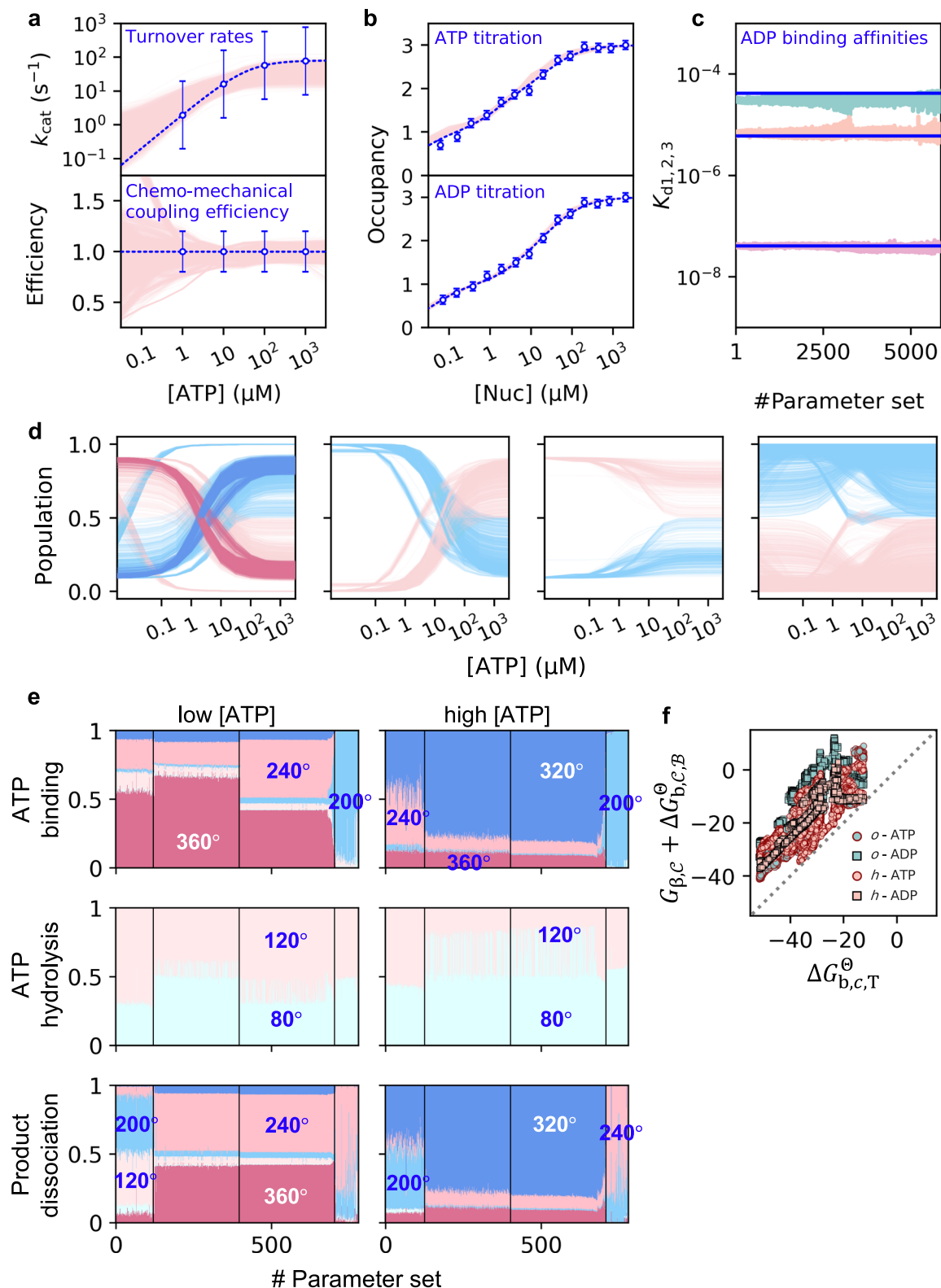
Based on the predicted probabilities (Fig. 2e, represented by the height of differently colored areas), we classified these parameter sets into four mechanistically distinct clusters (separated by vertical bars). However, none aligns with the consensus coupling scheme (disagreements summarized in Supplementary Table 3), for instance by predicting ATP hydrolysis at γ -orientations earlier than observed. Thus, the *ohc-s* variant ultimately fails this cross-validation test.

In summary, despite their large parameter spaces (14 free parameters), all three tested *ohc*-variants failed to explain all available experimental data. If true, this finding would suggest that the three-conformation hypothesis is fundamentally incompatible with the data, at least for these tested γ - β restrictions. However, before finally rejecting this hypothesis, we had to conclusively rule out two alternative explanations for this failure: insufficient parameter sampling and incorrect γ - β restrictions. We excluded the former because independent optimization runs from random starting points consistently converged to the same scenarios (Fig. 2e). The latter is addressed below, by systematically examining whether a viable choice was hidden among the other ~600 untested *ohc*-variants.

The three-conformation hypothesis is fundamentally incompatible with experimental data

Rather than exhaustively test the remaining *ohc*-variants via the computationally expensive training + cross-validation procedures, we first derived several physical constraints on the γ - β restrictions from the consensus coupling scheme to systematically reduce the number of candidates. These constraints are summarized in Supplementary Table 4 (see Methods and Supplementary Note 3 for detailed explanations). Briefly, this analysis leverages the fundamental binding-change principle^{6,15–17,35,68,73}: γ -subunit rotation induces sequential β -subunit conformational changes, altering their nucleotide binding affinities to drive the catalytic steps. Complemented by our earlier finding that a nucleotide-bound β -subunit energetically favors the closed conformation unless sterically forced open, this principle implies that γ -subunit rotation immediately following a catalytic step must modulate the accessible conformational ensemble of the β -subunit to induce the required conformational change.

Specifically, the constraint governing product dissociation at +320° relative to the ATP binding angle differentiates two families of model variants (Supplementary Equation (7) in Supplementary Table 4): Family A forces the β -subunit to fully open (*h* → *o*) during the +320° to +360° rotation, while Family B allows it to partially close (*o* → *h*). Thermodynamically, Family A implies a decrease in product



binding affinity upon this rotation, whereas Family B implies an increase (Supplementary Equations (41)–(42)). We excluded all Family B variants because an affinity increase contradicts the physical intuition that product dissociation is driven by reduced affinity^{6,8,15}, also supported by recent titration experiments observing a low-affinity site at 120°-dwells²². While Family B aligns with recent cryoEM structures³², this apparent discrepancy is resolved by re-interpreting the functional roles of these structures (Discussion).

Together with an additional plausible constraint for model parsimony (Condition *monotonic*, see Supplementary Table 5), these conditions (Supplementary Table 4) reduced the number of candidate *ohc*-variants to just three (Supplementary Table 8). However, all three were further excluded due to inconsistencies with other experimental data (Supplementary Table 8). Particularly, one of these is the previously tested *ohc-m* variant, which disagrees with the experimental apparent ADP binding affinities.

Fig. 2 | Training and cross-validation of the *ohc-s* variant. **a, b** Model predictions from the ensemble of trained parameter sets (pink curves) versus training data (blue circles), including turnover rates, chemo-mechanical coupling efficiency, and nucleotide titration curves. Vertical bars indicate the standard deviations of the likelihood functions (normal or log-normal distributions) adopted in the Bayesian training approach, as detailed in Supplementary Table 7. **c** Predicted apparent ADP binding affinities (green, orange and pink curves) versus the experimentally determined values (blue horizontal bars). **d** First cross-validation test: predicted populations of the 80°- (blue) and 120°-states (pink) versus ATP concentration ([ATP]). The trained parameter sets are classified into four groups based on their qualitative predictions. Only the leftmost group correctly reproduces the experimentally observed population shift from 120°-dwells at low [ATP] to 80°-dwells at

high [ATP]. 750 parameter sets are selected from this successful group (highlighted in darker blue and darker pink) and subjected to the second cross-validation test. **e** Second cross-validation test: predicted probabilities of the major catalytic steps (ATP binding, hydrolysis, product dissociation) at different γ -orientations for low (0.1 μ M) and high (1 mM) [ATP] are represented by the height of the differently colored areas in each panel. The selected 750 parameter sets are further classified into four mechanistically distinct clusters (separated by vertical bars). **f** A common property of all trained parameter sets: the free energy of a nucleotide-bound β -subunit is lowest when it adopts the catalytically active, closed *c* conformation. The plot compares the total free energies of an ATP/product-bound β -subunits in the *o* or *h* conformations (*y* axis) against that of an ATP-bound β -subunit in the *c* conformation (*x* axis). Source data are provided as a Source Data file.

This finding raises a deeper question: physicochemically, why are variants like *ohc-m* and *ohc-w* incompatible with the experimental apparent affinities? The key insight comes from recognizing that these apparent affinities are not properties of single, static conformations in our model. Instead, analytical derivations based on equilibrium approximation show that each apparent affinity emerges as a superposition—a Boltzmann-weighted ensemble average—of the microscopic binding affinities of all accessible conformations dictated by the γ - β restrictions (Methods, see particularly Eq. (19)). Thus, for our model to reproduce the vastly different experimental apparent affinities²¹, the γ - β restrictions should impose different conformational ensembles onto the three β -subunits. Nevertheless, the *ohc-w* variant has identical γ - β restrictions for two β -subunits, inherently leading to two similar apparent affinities (Supplementary Fig. 1c) that contradict the experimental values. Similarly, although mathematically more complex, the *ohc-m* variant is also incompatible with the experimental apparent affinities (Supplementary Note 5).

Taken together, this systematic analysis considering physical-chemical constraints demonstrates that no *ohc*-variant can simultaneously explain all available experimental data, strongly suggesting that the underlying three-conformation hypothesis is insufficient.

Several *ohc*₁*c*₂-variants are found to reproduce all available experimental data

We were therefore prompted to consider the next simplest extension of our model: including a fourth β -subunit conformation, resulting in *ohc*₁*c*₂-variants featuring two distinct closed states, *c*₁ and *c*₂ (Fig. 1d), tentatively associated with the experimentally observed β_{TP} and β_{DP} structures^{4,32,65}. While the combinatorial possibilities of γ - β restrictions initially yield over 14,000 *ohc*₁*c*₂-variants (Methods), we systematically reduced this number to fifteen candidates (Supplementary Note 4), applying the constraints derived from the consensus chemo-mechanical coupling scheme (Supplementary Table 4), supplemented by a few additional constraints based on observed structures and model parsimony (Supplementary Table 5).

As the five simplest and most plausible candidates, we selected three *ohc*₁*c*₂- and two *ohc*₁*c*₂^{*}-variants, and applied the training + cross-validation procedures (Methods) to conclusively determine if any *ohc*₁*c*₂-variant could explain all available data (Table 3). All five candidates successfully reproduced both the training and initial validation datasets (Supplementary Figs. 3–5). Interestingly, these five variants predicted two distinct mechanistic scenarios regarding the γ -orientations of the major catalytic steps. The *ohc*₁*c*₂^{*}-*w* variant predicted these γ -orientations independent of [ATP] (Scenario 1; Fig. 3a), whereas the other four variants predicted ATP-dependent γ -orientations (Scenario 2; e.g., Fig. 3b, Supplementary Fig. 5b–d). For instance, in Scenario 2, product dissociation shifts to an earlier orientation (+240°) at low [ATP], while ATP binding occurs immediately after product dissociation (+320°) at high [ATP]. Both scenarios, however, remain compatible with the consensus chemo-mechanical coupling scheme.

As an additional cross-validation test, in Fig. 3c, we compared the predicted ratios of bound ADP to ATP ($\nu_D : \nu_T$) against data from an early tryptophan fluorescence experiment²⁰. The model predictions (pink curves) were tested against the experimental observation (black dots, adapted from Fig. 6 in ref. 20) that $\nu_D : \nu_T \approx 2 : 1$ at saturating [ATP].

The variant *ohc*₁*c*₂^{*}-*m*.1 predicted $\nu_D : \nu_T$ below or around 1:1 at high [ATP] (second panel), and thus was excluded. In contrast, the other four variants reproduced this observed 2:1 ratio. Therefore, this cross-validation test leaves these four variants (*ohc*₁*c*₂^{*}-*w*, *ohc*₁*c*₂^{*}-*m*.1, *ohc*₁*c*₂^{*}-*m*.2, and *ohc*₁*c*₂^{*}-*m*.2) as candidates for a minimal model of F₁-ATPase.

The candidate variants also reproduce observed single-molecule trajectories

We finally tested if the four candidate *ohc*₁*c*₂-variants also agree with the single-molecule experiments where γ -subunit rotation was tracked over time using attached probes like actin filaments or colloidal gold beads^{8,13,14}. To this end, we performed kinetic Monte-Carlo (KMC) simulations of these variants, closely resembling the experiments.

As depicted in Fig. 4a, the probe (e.g., a bead, yellow) acts as an external load coupled via a linker (modeled as a harmonic spring with force constant κ) to the γ -subunit (green). Following the experiments, we tracked the circular motion of the probe centroid, rather than the γ -subunit itself. Given the probe's large size relative to F₁-ATPase, its motion in the solvent was assumed to be overdamped and thus described by an overdamped Langevin equation (OLE, (Eq. (21)) with rotational friction drag coefficient, ξ . We estimated $\kappa \approx 5 k_B T$ and $\xi \approx 6.7 \times 10^{-5} k_B T \cdot s$, based on experimental reports^{8,13,14}. By coupling the KMC simulation of our Markov model with numerical integration of the OLE (Methods), we generated trajectories of the angular positions of the probe centroid comparable to those measured experimentally.

The simulations of the variant *ohc*₁*c*₂^{*} are reported in Fig. 4b–g. Figure 4b shows sample trajectories for [ATP] ranging from 1 μ M to 1 mM. These trajectories closely resemble those recorded experimentally (cf., e.g., Fig. 4 in ref. 8), notably reproducing the characteristic 80° and 40° rotational substeps. Furthermore, the dependencies of rotational rate on viscous friction and [ATP] (Fig. 4c, d) agree well with the single-molecule measurements (cf. Figs. 2 and 3 in ref. 8). The other three candidate variants produced similar trajectories and dependencies that also match the experimental observations.

A more detailed quantitative analysis via a hidden Markov model (Methods) of the simulated trajectories of the *ohc*₁*c*₂^{*}-*w* variant yielded dwell-time distributions (Fig. 4e–g) consistent with ATP binding limiting the 120° dwells and product dissociation limiting the 80° dwells, again matching the experimental conclusions^{8,15}. Given that all four candidate variants share similar chemo-mechanical coupling schemes (Fig. 3 and Supplementary Fig. 5), simulated trajectories, and dependencies, we expect them to also share similar dwell-time distributions.

Table 3 | Definition and evaluation of the five most plausible ohc_1c_2 -variants

Variant	γ - β restrictions (Γ_k^β)				Number of states	Training +Cross-validation tests	Cross-validation test 3
$ohc_1^*c_2-w$	ϕ	β_1	β_2	β_3	972	Succeeded	Succeeded
	80°	o, h	o, h, c_1, c_2	o, h, c_2			
	120°	o	o, h, c_2	o, h, c_1, c_2			
$ohc_1^*c_2-m.1$	ϕ	β_1	β_2	β_3	864	Succeeded	Failed
	80°	o, h	o, h, c_1, c_2	o, h, c_2			
	120°	o	o, h	o, h, c_1, c_2			
$ohc_1^*c_2^*-m.1$	ϕ	β_1	β_2	β_3	864	Succeeded	Succeeded
	80°	o, h	o, h, c_1, c_2	o, h, c_2			
	120°	o	o, h	o, h, c_1, c_2			
$ohc_1^*c_2-m.2$	ϕ	β_1	β_2	β_3	864	Succeeded	Succeeded
	80°	o, h	o, h, c_1	o, h, c_1, c_2			
	120°	o	o, h	o, h, c_1, c_2			
$ohc_1^*c_2^*-m.2$	ϕ	β_1	β_2	β_3	864	Succeeded	Succeeded
	80°	o, h	o, h, c_1	o, h, c_1, c_2			
	120°	o	o, h	o, h, c_1, c_2			

For each variant, the table lists its defining γ - β restrictions, the number of resulting Markov states, and the outcome of the training and cross-validation procedures (Methods). Training: A Bayesian approach was used to find parameter sets that reproduce the training data. Cross-validation: The trained parameter sets were tested against three sets of validation data, first (cross-validation test 1) the observed population shift between dwells, second (cross-validation test 2) the consensus chemo-mechanical coupling scheme (Table 6 and Supplementary Fig. 2), and third (cross-validation test 2) an additional test against the measured ADP:ATP occupancy ratio. “Succeeded” or “Failed” indicates whether parameter sets consistent with the respective data were found.

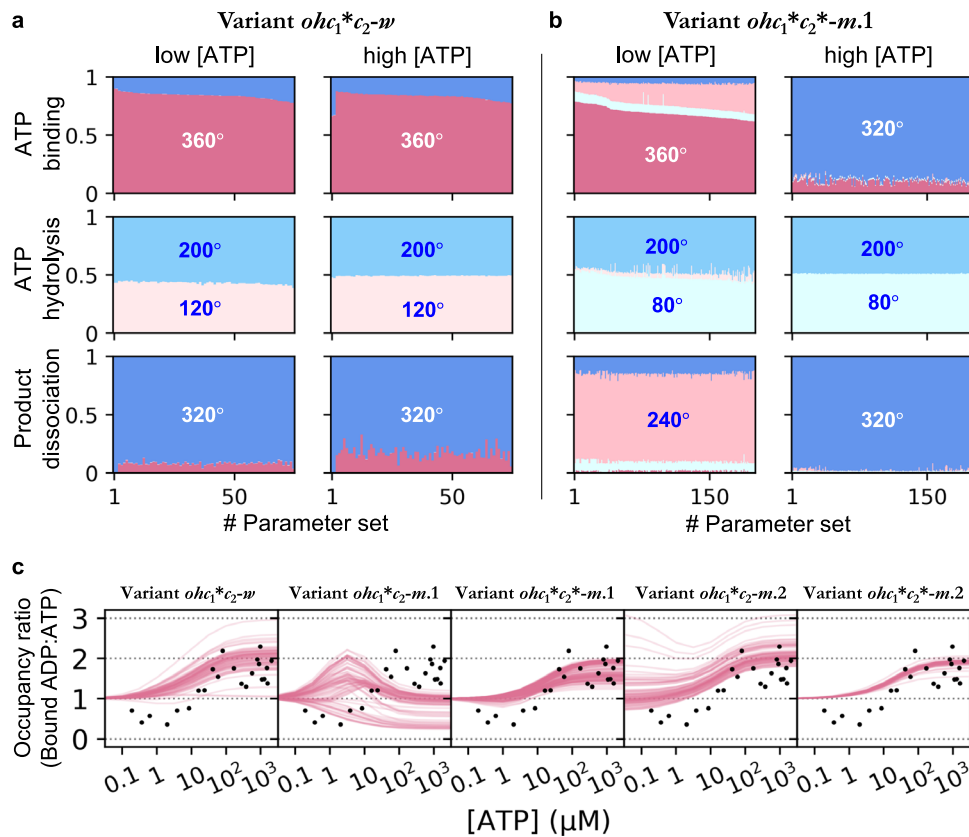


Fig. 3 | Cross-validation of the ohc_1c_2 -variants. a, b Second cross-validation test: predicted probabilities of the major catalytic steps at different γ -orientations for two representative variants, $ohc_1^*c_2-w$ in (a) and $ohc_1^*c_2^*-m.2$ in (b). The representation is consistent with Fig. 2e. **c** Third cross-validation

test: the predicted ratio of bound ADP to ATP (pink curves) for all five tested ohc_1c_2 -variants, compared against the validation data from tryptophan fluorescence experiments (black dots). Source data are provided as a Source Data file.

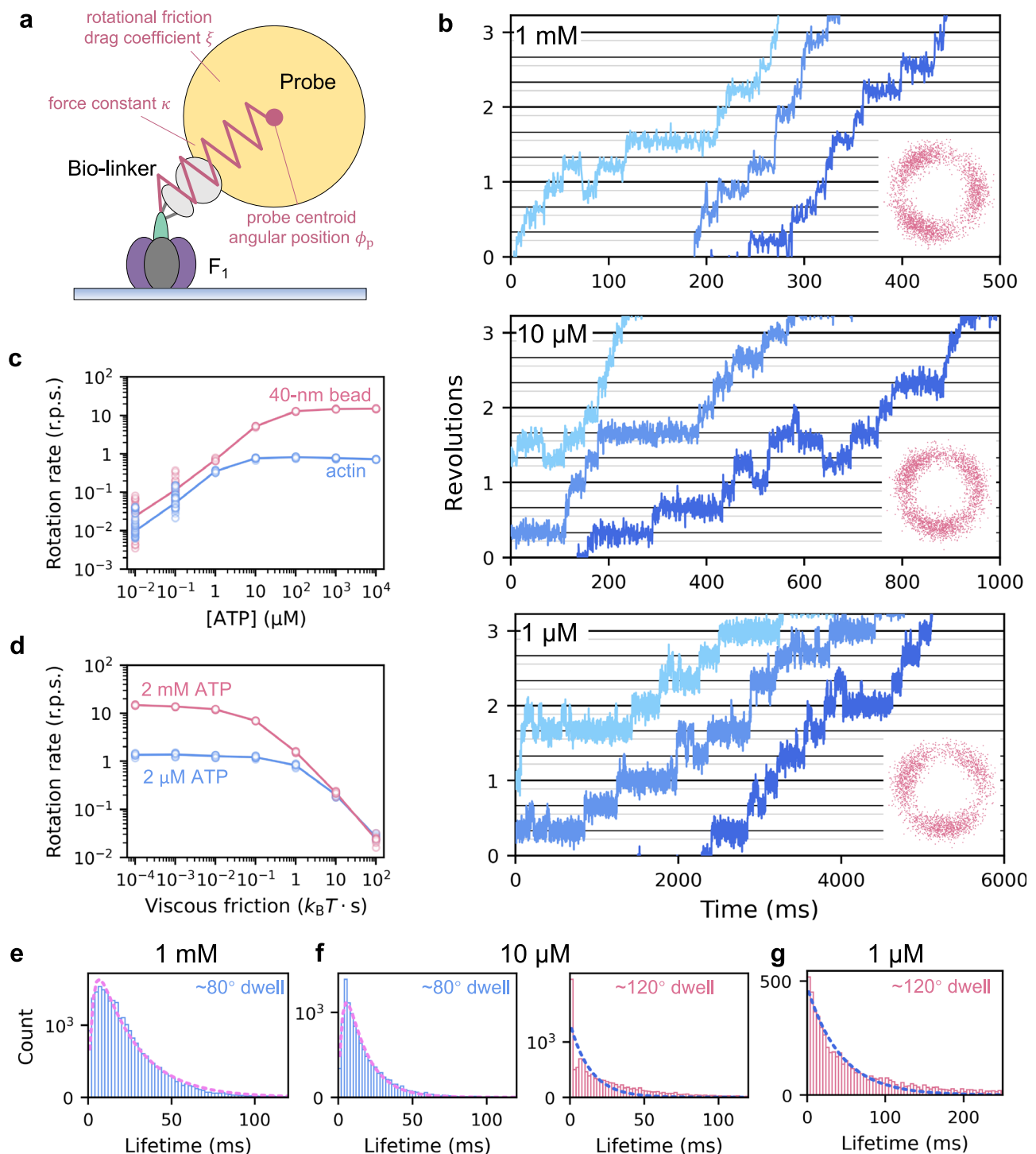


Fig. 4 | Kinetic Monte-Carlo simulations of the *ohc1c2-w* variant reproduce single-molecule observations. **a** Kinetic Monte-Carlo (KMC) simulation setup mimicking single-molecule experiments, where a probe (yellow) is coupled to the γ -subunit (green) via a bio-linker (force constant κ), and is subjected to rotational friction drag coefficient ξ . Adapted from ref. 8 (cf. Figure 1b). **b** Sample KMC trajectories of the probe rotation at 1 mM, 10 μ M, and 1 μ M ATP. In each panel, the three curves are continuous; later curves are shifted to save space. The gray horizontal lines are placed 40° below the black lines. Insets: angular positions of the probe taken from the sample trajectories and randomly placed around the unit cycle. The simulations qualitatively reproduce the experimentally observed discrete 80° - and 40° -substeps. **c** Predicted rotational rate versus ATP concentration

for two loads: a 40-nm gold colloidal bead (pink), and a 1- μ m actin (blue). **d** Predicted rotational rate versus viscous friction at high (2 mM, pink) and low ATP concentrations (2 μ M, blue). **c, d** Circles represent rates calculated from individual trajectories; solid curves are averages. Both **c, d** show good quantitative agreement with experimental measurements (cf. Figs. 2 and 3 in ref. 8). **e–g** Dwell-time distributions (histograms) extracted from the simulated trajectories via Hidden Markov Analysis (Methods). **e** At 1 mM ATP, only $\sim 80^\circ$ -dwells are identified. **f** At 10 μ M ATP, both $\sim 80^\circ$ -dwells and $\sim 120^\circ$ -dwells are identified. **g** At 1 μ M ATP, only $\sim 120^\circ$ -dwells are identified. The dwell-time distributions for the 80° - and 120° -dwells are fitted by double and single exponential functions, respectively (violet and blue dashed curves). Source data are provided as a Source Data file.

This overall consistency provides further cross-validation for all four candidate variants.

Discussion

Aiming at a minimal yet thermodynamically consistent model for F_1 -ATPases from two bacterial species (*E. coli* and *Bacillus PS3*), here we have developed a chemo-mechanical Markov model that integrates conformational and chemical DOFs, as well as steric interactions between the γ - and β -subunits. We used a Bayesian training approach to infer the free energies and free energy barriers from the relevant experimental data extracted from literature^{8,19,21,75}. By systematic model comparison via our training + cross-validation strategy, we tested competing mechanistic hypotheses, particularly the three-conformation versus four-conformation hypotheses, and eventually identified several model variants with four distinct conformations of individual β -subunits (*ohc*₁*c*₂-variants) which all agree with and predict all relevant experimental data (Methods). These resulting candidates provide a coherent thermodynamic and kinetic framework that quantitatively explains the data for these $\sim 80^\circ/\sim 40^\circ$ sub-stepping enzymes.

Our study reconciles the decade-long controversy over whether a bi-site or a tri-site mechanism best describes ATP hydrolysis at full activity. In contrast to previous binding change models^{5,8,20,22,31–35}, which assume only one or at most very few catalytic pathways, our model involves by construction many parallel catalytic pathways, and predicts these to contribute to the overall activity with different fluxes that change with ATP concentration (Supplementary Fig. 6). At physiological (millimolar) ATP concentration, pathways resembling the tri-site mechanism^{40,52} dominate, wherein the total nucleotide occupancy alternates between three and transiently two. At micromolar ATP concentration, in contrast, bi-site pathways dominate, wherein the total occupancy alternates between one and transiently two. Accordingly, the total occupancy gradually decreases from three to one with decreasing ATP concentration, agreeing with published ATP titration data^{19,21,66,67}.

Strikingly, we found no model variant with only three functionally distinct conformational states of each β -subunit (e.g., open, half-open, closed) that agrees with all relevant experimental data. We therefore conclude that a minimum of four functional states is required—and, from the above findings, also suffices—for proper F_1 -ATPase function. Indeed, it was proposed early on that a tri-site mechanism requires at least four conformations⁶⁷; furthermore, recent tryptophan fluorescence experiments²² suggested the existence of at least four conformations with distinct nucleotide binding affinities. Particularly, several authors have postulated at least two different closed conformations, based on the observed structural and thermodynamic asymmetry of the three β -subunits^{4,5,71}, in line with our *ohc*₁*c*₂-variants. This idea is also supported by a principal component analysis of all available β -subunit structures (Supplementary Fig. 7), demonstrating considerable heterogeneity of those structures classified as closed.

Helping to resolve another debate^{46,48,58}, our study strongly supports the notion that for a nucleotide-bound β -subunit, the catalytically active, closed conformation is energetically more favorable than the open and partially open conformations. In fact, according to our Markov model, this energetic property of the β -subunits is a necessary condition for establishing tight coupling between catalysis and rotation (see Fig. 2f and Supplementary Note 1). A plausible explanation is that under this condition, a β -subunit is unlikely to open spontaneously and release the product after ATP hydrolysis, unless forced by the γ -subunit, thereby ensuring tight chemo-mechanical coupling. Conversely, if the closed conformation for a nucleotide-bound β -subunit were not the most stable one, the β -subunits could go through catalytic cycles spontaneously and independently of the γ -subunit orientation, resulting in a catalytic rate higher than the rotational rate and a correspondingly reduced coupling efficiency. Notably, this

energetic property was observed independently in previous atomistic simulation studies^{46,58}, where the β -subunit that was initially kept open by the bulge of the γ -subunit closed spontaneously after removing the bulge through enforced rotation of the γ -subunit.

Our model quantitatively reconciles the classical binding change paradigm with the inherent stochastic nature of rotary catalysis. In line with Arai et al.²⁹, our model by construction conceptualizes the γ -subunit rotation as random thermal fluctuations which are neither directly nor deterministically coupled to ATP hydrolysis (Supplementary Table 2). In this sense, our model represents a Brownian-ratchet mechanism, which is able to explain the complex and intriguing stochastic rotational behaviors of the central stalk at low ATP concentrations observed in single-molecule experiments^{8,15,52}. First, in our KMC simulations, the γ -subunit often rotates back and forth stochastically over several 120° steps, directly mirroring the irregular rotations observed experimentally^{15,52}. Remarkably, when no nucleotide is bound within the three catalytic sites, the simulated irregular rotations can accumulate up to several revolutions. This prediction directly supports the hypothesis that the irregular rotations represent “Brownian hopping among the three equivalent orientations in a nucleotide-free F_1 -ATPase”⁵². Second, our model predicts that one ATP hydrolysis event can trigger larger than 120° rotations, explaining the experimental measurements somewhat misleadingly termed over 100% coupling efficiencies⁸. Notably, these events do not violate energy conservation; they occur only under sufficiently low external load, meaning the work performed even over several 120° steps remains smaller than the free energy released by one ATP hydrolysis. Rather, these results are perfectly in line with a Brownian-ratchet mechanism^{29,30}. Specifically, at low ATP concentrations where the nucleotide occupancy is only one or less, large rotational fluctuations occur almost uncoupled from ATP hydrolysis. As nucleotide exchange alters the affinity landscape of the catalytic subunits, these stochastic fluctuations are directionally rectified, allowing one hydrolysis event to trigger a full revolution in the absence of mechanical load.

With increasing ATP concentration and nucleotide occupancy, the rotation becomes more tightly coupled to ATP hydrolysis. Irregular rotations rarely occur, and the coupling efficiency approaches 100% (Fig. 2 and Supplementary Figs. 3 and 4). In this high-ATP-concentration regime, the macroscopic behavior of the F_1 -ATPase becomes more similar in spirit to the deterministic picture as in classical binding change models, functioning apparently like a power-stroke. Taken together, the agreement of our model with multiscale experimental data—including both the microscopic, stochastic single-molecule observations^{8,15,52} and the macroscopic, deterministic measurements^{19,21}—demonstrates that the classical binding change paradigm is fully compatible with the microscopic stochasticity of this molecular machine, which underscores that an intrinsic Brownian-ratchet mechanism is both feasible and sufficient for rotary catalysis.

Although the four candidate models agree with all relevant experimental data, and although they share many other properties, they do differ in mechanistic details (Table 4). An example is which of the two closed β -subunits in the classic crystal structure nomenclature⁴, β_{TP} or β_{DP} , is the high affinity site (third column); here, the experimental evidence is still inconclusive, with some results pointing to β_{TP} ^{31,72,80} and others to β_{DP} ^{4,5}. Similarly, the number of β -subunits that can be simultaneously catalytically active varies between one and two among the four variants (fourth column); experimental evidence supporting both possibilities has been reported^{5,71,72}. Therefore, based on these current results, we cannot conclude which of these four candidates is more or less likely. As of now, it seems to us that the available experimental data does not allow us to distinguish between these different mechanistic details.

It also remains a challenge to interpret the functional roles of experimentally resolved atomistic structures within the F_1 -ATPase catalytic cycle^{63,70,81–83}, particularly given the differing cryoEM

Table 4 | Comparison of the four candidate models

Variant	Representative functional states of the catalytic (80°) and ATP-waiting (120°) dwells	High-affinity site	Catalytically active β -subunit(s) in the catalytic dwell
$ohc_1^+c_2^-w$	80°: hc_1c_2 , oc_1c_2 ; 120°: oc_2c_1	$\beta_{DP} (c_1^+)$	Only β_{DP}
$ohc_1^+c_2^+-m.1$	80°: hc_1c_2 , oc_1c_2 ; 120°: ohc_1 , ooc_1	$\beta_{DP} (c_1^+)$	Both β_{TP} and β_{DP}
$ohc_1^+c_2^-m.2$	80°: hc_1c_2 , oc_1c_2 ; 120°: ohc_2 , ooc_2	$\beta_{TP} (c_2^-)$	Both β_{TP} and β_{DP} , although β_{TP} energetically favors the inactive c_2^- conformation
$ohc_1^+c_2^+-m.2$	80°: hc_1c_2 , oc_1c_2 ; 120°: ohc_2 , ooc_2	$\beta_{TP} (c_2^+)$	Both β_{TP} and β_{DP}

structures reported by Nakano et al.⁶⁵ and Sobti et al.³² for the ~80° catalytic and ~120° ATP-waiting dwells (Fig. 9c–f). Whereas these structures share similar nomenclatures fully compatible with single-molecule FRET measurements⁷⁰, there is a key difference regarding the conformational progression of the β_1 -subunit upon the 80° → 120° rotation: does it open further (as suggested by Nakano et al.⁶⁵, Supplementary Fig. 9c, d), or close partially (as suggested by Sobti et al.³², Supplementary Fig. 9e, f)? Our model provides a thermodynamic framework to resolve this ambiguity by mapping these structures to distinct functional stages within the catalytic cycle. Specifically, for the joint functional states of the three β -subunits in the 80°- and 120°-dwells, our model predicts that the most populated pathway involves the β_1 -subunit opening ($h \rightarrow o$) and a decrease in nucleotide binding affinity to facilitate product dissociation^{6,15,22,70,73} (Table 4, second column; Supplementary Figs. 9 and 10a, b). Along this pathway, the 80°-dwell is most populated by an *hcc* state (Supplementary Fig. 9a), representing the stage where the β_1 -subunit awaits product dissociation. Upon rotation to 120°, the enzyme converts to an *ohc* state (Supplementary Fig. 9b), representing the ATP-waiting stage. We note that in this context, the oc_2c_1 state predicted by the $ohc_1^+c_2^-w$ variant can also be interpreted as an *ohc*-like state (Supplementary Note 7).

Notably, the Nakano structures (Supplementary Fig. 9c, d) align well with this predicted progression: despite both being labeled open (β_o), the catalytic interface of the β_1 -subunit appears more compact at 80° than at 120°. Therefore, we suggest to assign the Nakano structures to these predicted primary functional stages (*hcc* and *ohc*). In contrast, the Sobti structures (Supplementary Fig. 9e, f) suggest an opposite $o \rightarrow h$ progression: here, the β_1 -subunit structure is less compact at 80° than at 120° (β_o versus β_{HC}). However, as demonstrated by our analysis of Family B variants which predict such a progression as the dominant pathway (Results), this progression would imply an affinity increase upon the 80° → 120° rotation, in contrast to what has been measured previously²², as well as to analysis of single-molecule data^{8,15}. Furthermore, several structures resembling the Nakano *ohc*-like structure (Supplementary Fig. 9d) have been reported for various modified F₁-ATPase constructs in the ATP-waiting dwell^{64,78,79}, featuring a β_1 -subunit consistently more open than the Sobti β_{HC} structure at 120° (Supplementary Fig. 9f). Given these discrepancies, we suggest that the Sobti structures likely capture different functional stages. For the 80°-dwell, we would assign their structure (Supplementary Fig. 9e) rather to the *occ* state of our model (Supplementary Fig. 9a), which is transient in the native cycle, appearing only after product dissociation in the β_1 -subunit but before ATP hydrolysis completes in the β_2 -subunit. This assignment raises the question of why this transient structure was actually seen in the cryoEM sample, which can be explained by the use of a mutant that intentionally slowed down the hydrolysis step³², thereby sufficiently stabilizing this structure. Similarly, for the 120°-dwell, the Sobti structure (Supplementary Fig. 9f) might represent an intermediate captured shortly after ATP binding, as the β_1 -subunit begins to close.

Taken together, our model integrates these differing structures into a unified mechanistic picture. For the 80°-dwell, the Nakano

hcc-like (Supplementary Fig. 9c) and the Sobti *occ*-like structures (Supplementary Fig. 9e) likely correspond to the pre- and post-product-dissociation stages, respectively, consistent with the observation of two distinct rate-limiting steps in single-molecule experiments^{8,15}. For the 120°-dwell, the Nakano *ohc*-like (Supplementary Fig. 9d) and Sobti *hcc*-like structures (Supplementary Fig. 9f) likely represent the pre- and post-ATP-binding stages, respectively. This interpretation requires the involved structures to be rather flexible, which is in line with previously observed large fluctuations of the γ -subunit within this dwell^{15,81}.

Striking the best balance between model complexity and sufficiently few model parameters is a notorious challenge. Building upon our minimal model, several routes exist for future refinement and extension. For example, our minimal model assumes that the steric repulsions between the γ - and β -subunits are the main (and only) determinants for efficient chemo-mechanical coupling. It is known, however, that weaker effects such as allosteric communication within the $\alpha_3\beta_3$ ring or finer details of the γ - β interaction landscape (e.g., electrostatics) also contribute and possibly modify the mechanism of the chemo-mechanical energy transduction. One line of evidence is the structural asymmetry and residual activity of γ -depleted and γ -truncated F₁-ATPase constructs^{76–79}. It would therefore be a tempting future extension of our model to also include energy terms—of course, requiring additional parameters—which describe these weaker and, in our view, secondary effects. Other possible extensions include explicitly modeling phosphate to dissect the timing and kinetics of its release; testing the model against a wider range of experimental data, such as those from mutant F₁-ATPase^{15,34}, and potentially moving beyond binding change models to consider continuous rotation of the γ -subunit.

While our minimal Markov model was developed specifically for bacterial F₁-ATPases, the underlying Brownian-ratchet picture can serve as a universal principle for chemo-mechanical energy transduction across the entire rotary ATPase superfamily, including F-ATPase from other species^{61–63} as well as V- and A-ATPases^{23–30}. For the specific example of bacterial F-ATPase, our study demonstrates how this general principle can be implemented in quantitative and thermodynamically consistent terms. Similarly, for each of the other rotary ATPases, a fine-tuned and high-dimensional free energy landscape underlies and supports this principle, which, in our model, is explicitly quantified by the parameters, including free energies and activation barriers.

While the fundamental energy transduction principle is shared, these rotary ATPases differ significantly in subunit complexity, regulatory mechanisms, and rotational sub-stepping behaviors^{23–30}. These differences imply that the underlying energy landscapes are fine-tuned differently by evolution, and, accordingly, respective Markov models will be defined by different free energy levels and barrier heights. Determining these will require similarly comprehensive measurements as were available for F-ATPase, and it will be exciting to compare the obtained different free energy landscapes to further extract common fundamental principles as well as differences between families and

between species, thereby revealing how evolution has tuned their stabilities, binding affinities, catalytic rates, and coupling efficiencies to optimally adapt their functional mechanisms to the respective biological demands.

Beyond rotary ATPases, the theoretical framework established in this study for developing thermodynamically consistent Markov models is very general and therefore can be applied to a broad range of macromolecular machines that combine conformational motions with ligand binding/unbinding or enzymatic reactions to achieve their function. Particularly, the use of thermodynamically consistent Markov models, which include all relevant states as well as conformational and catalytic pathways, should help to overcome the limitations and sometimes misleading interpretations of simpler, Michaelis–Menten-like descriptions that consider only a few presumed pathways and apparent binding affinities. For more complex and conformationally coupled enzyme dynamics, the latter may be insufficient, as testified by the bi-site versus tri-site controversy.

Finally, the systematic, data-driven model comparison approach demonstrated here also serves to assess competing mechanistic hypotheses against available experimental data. An essential aspect of this approach is a training + cross-validation strategy, which, as detailed in Methods, renders complex systems computationally tractable while ensuring thermodynamic consistency.

Methods

Mathematical formulation of the chemo-mechanical Markov model

Here, we present the detailed mathematical formulation for our chemo-mechanical Markov model, including the definitions of the Markov state space, the free energies of the Markov states, and the transition rates, as well as the master equation governing the time evolution of the system.

Each chemo-mechanical state (Markov state) $\mathbf{s}$ in our model is distinctly defined by a seven-dimensional vector specifying the values of the seven DOFs introduced previously (Table 1):

$$\mathbf{s} = (\phi, C^{(1)}, C^{(2)}, C^{(3)}, B^{(1)}, B^{(2)}, B^{(3)}), \quad (1)$$

where ϕ is the γ -subunit orientation, $C^{(k)}$ is the conformation of the k -th β -subunit, and $B^{(k)}$ is its nucleotide binding state ($k = 1, 2, 3$). The complete set of Markov states includes all combinatorially possible combinations of these seven DOFs. Formally, this state space is constructed by the Cartesian product of the seven DOFs.

For each Markov state, its free energy (state free energy) is decomposed into a sum of the conformational free energies of individual β -subunits, their binding free energies, and the interaction energies between the γ - and β -subunits:

$$G(\mathbf{s}; c_T, c_D) = \sum_{k=1,2,3} [G_{\beta,C^{(k)}} + \Delta G_b(C^{(k)}, B^{(k)}; c_T, c_D) + G_{\gamma\beta}(\phi, C^{(k)})] \quad (2)$$

Here, a free energy $G_{\beta,C}$ is assigned to each β -subunit conformation ($C = o, h, c$ for *ohc*-variants; $C = o, h, c_1, c_2$ for *ohc*₁₂-variants), and is defined as a model parameter (conformational free energies, Table 5).

The binding free energy, $\Delta G_b(C^{(k)}, B^{(k)}; c_T, c_D)$, which depends on the bulk nucleotide concentrations c_T and c_D , is defined by:

$$\Delta G_b(C^{(k)}, B^{(k)}; c_T, c_D) = \begin{cases} \Delta G_{b,C^{(k)},B^{(k)}}^0 - k_B T \ln c_{B^{(k)}}, & B^{(k)} = T, D; \\ 0, & B^{(k)} = E. \end{cases} \quad (3)$$

Here, two free energies $\Delta G_{b,C,B}^0$ ($B = T, D$) are assigned to each β -subunit conformation for ATP and ADP binding, respectively, under

the standard condition of 1 mM ATP/ADP. They are also defined as model parameters (standard ATP/ADP binding free energies, Table 5).

The γ - β interaction energy $G_{\gamma\beta}(\phi, C^{(k)})$ primarily models the steric repulsions between the γ - and β -subunits. As introduced in Results, we use a strong simplification, assuming an infinitely high energy for those sterically disfavored conformations, thus precluding them from being populated; the remaining conformations are thermodynamically accessible. The hyperparameter γ - β restrictions, also introduced previously, specify the set of accessible conformations, Γ_k^ϕ , for each β -subunit (β_k), at each γ -orientation (ϕ). Formally, this interaction energy $G_{\gamma\beta}(\phi, C^{(k)})$ is defined by:

$$G_{\gamma\beta}(\phi, C^{(k)}) = \begin{cases} +\infty, & C^{(k)} \notin \Gamma_k^\phi; \\ 0, & C^{(k)} \in \Gamma_k^\phi. \end{cases} \quad (4)$$

Additionally, for Family A variants (Supplementary Equation (7) in Supplementary Table 4), a slightly favorable interaction energy of $-2.3 k_B T$ is assumed for the β_k -subunit adopting the *o* conformation at γ -orientation $\phi = 120^\circ \times k$ ($G_{\gamma\beta}(\phi, C^{(k)}) = -2.3 k_B T$ if $\phi = 120^\circ \times k$ and $C^{(k)} = o$), describing the $\sim 120^\circ$ ATP-waiting dwell observed in single-molecule experiments^{8,15}.

The transition rates for all elementary transitions included in our model are defined in Supplementary Table 2. These definitions are grounded in transition state theory and the principle of local detailed balance, as outlined below.

The transition rate from a Markov state i to another Markov state j in our model, denoted $r_{i \rightarrow j}$, is determined by the corresponding rate coefficient $k_{i \rightarrow j}$ and the molecularity (reaction order) of the process. According to transition state theory, the rate coefficient $k_{i \rightarrow j}$ is determined by the free energy barrier $\Delta G_{i \rightarrow j}^\ddagger$:

$$k_{i \rightarrow j} = f_{\text{att}} \exp\left(-\frac{\Delta G_{i \rightarrow j}^\ddagger}{k_B T}\right) \quad (5)$$

Here, f_{att} is a constant, the attempt frequency. It serves as a global pre-exponential factor; its specific value (here chosen to be 10^9 s^{-1}) is not critical, as any different choice would be absorbed by a uniform shift in the fitted free energy barriers.

To enforce the principle of local detailed balance for every elementary transition (Results), the backward rate coefficient $k_{j \rightarrow i}$ (from state j to state i) is related to $k_{i \rightarrow j}$ by:

$$k_{j \rightarrow i} = k_{i \rightarrow j} \exp\left(\frac{G_j - G_i}{k_B T}\right), \quad (6)$$

where G_i, G_j are the free energies of the two states.

For a uni-molecular (first-order) transition, e.g., rotation of the γ -subunit, $k_{i \rightarrow j}$ defined above is a first-order rate coefficient, while the transition rate $r_{i \rightarrow j} = k_{i \rightarrow j}$. For a bi-molecular (second-order) transition, e.g., nucleotide binding, $k_{i \rightarrow j}$ defined above is a second-order rate coefficient, while $r_{i \rightarrow j}$ is the product of the $k_{i \rightarrow j}$ and the nucleotide concentration. Note that, because our model describes the internal dynamics of a single F₁-ATPase molecule, $r_{i \rightarrow j}$ does not need to include its own concentration.

Notably, as demonstrated in Supplementary Table 2, to define the large number of transition rates without introducing an excessive amount of free parameters, we made simplifying assumptions about the energy barriers. For example, we assume a uniform free energy barrier $\Delta G_\beta^\ddagger$ governs all β -subunit conformational transitions (Supplementary Equation (1) in Supplementary Table 2), limiting their maximum rate to r_β^{max} .

Another important assumption is that the rates of ATP hydrolysis and synthesis in a catalytically active β -subunit are equal. This assumption is based on experimental evidence suggesting that the hydrolysis and synthesis reactions are approximately in equilibrium

Table 5 | Model parameters

Symbol	Description	Notes
$\{G_{\beta,c}\}$	Conformational free energies	For <i>ohc</i> -variants, $G_{\beta,c} = 0$ is defined as the zero reference; for <i>ohc</i> ₁ <i>c</i> ₂ -variants, $G_{\beta,c_1} = 0$ is defined as the zero reference.
$\{\Delta G_{b,c,B}^0\}$	Standard ATP/ADP binding free energies	For all catalytically active conformation(s), $\Delta G_{b,c,T}^0$ and $\Delta G_{b,c,D}^0$ are subject to constraint by thermodynamic consistency (Eq. (7)).
$\{\Delta G_{b,c,B}^{\ddagger}\}$	Free energy barriers of ATP/ADP binding	N/A
$\Delta G_{\gamma}^{\ddagger}$	Free energy barrier of γ -subunit rotation	Limit the maximum rate for all $\pm 40^\circ$ and $\pm 80^\circ$ substeps
$\Delta G_{\beta}^{\ddagger}$	Free energy barrier of β -subunit conformational transition	Limit the maximum rate for all conformational transitions ($o \leftrightarrow h$, $h \leftrightarrow c$, $h \leftrightarrow c_1$, $h \leftrightarrow c_2$, $c_1 \leftrightarrow c_2$)
$\Delta G_{chem}^{\ddagger}$	Free energy barrier of ATP hydrolysis/synthesis in any catalytically active β -subunit	Assumed symmetric for both hydrolysis and synthesis

The β -subunit conformation $C = o, h, c$ (*ohc*-variants) or $C = o, h, c_1, c_2$ (*ohc*₁*c*₂-variants). The binding state $B = T, D$ for ATP and ADP, respectively. The catalytically active conformations refer to c for *ohc*-variants, c_1 for *ohc*₁*c*₂-variants, and both c_1 and c_2 for *ohc*₁*c*₂-variants.

within the catalytic site^{6,66–68}. To ensure overall thermodynamic consistency, this assumption implies that the difference between the standard ATP and ADP binding free energies must exactly compensate for the free energy of ATP hydrolysis in solution. We therefore enforce this condition by imposing a corresponding constraint on the standard binding free energies: for any catalytically active conformation C ,

$$\Delta G_{b,c,T}^0 - \Delta G_{b,c,D}^0 = -30.5 \text{ kJ/mol}, \quad (7)$$

where -30.5 kJ/mol is the standard free energy of ATP hydrolysis in solution⁸⁴.

These transition rates are used to construct the transition rate matrix, $\mathbf{R}$. The time evolution of our Markov model, modeled as a continuous time Markov process, is described by the master equation:

$$\frac{d\mathbf{p}}{dt} = \mathbf{R}(\boldsymbol{\Omega}; c_T, c_D)\mathbf{p}, \quad (8)$$

where $\mathbf{p}$ is the vector of the populations of the Markov states. The off-diagonal element of $\mathbf{R}$ represents the transition rate from state j to state i : $R_{ij} = r_{j \rightarrow i} (i \neq j)$; the diagonal element $R_{ii} = -\sum_{j \neq i} r_{i \rightarrow j} \equiv r_i$ is the total rate leaving state i . The transition rates R_{ij} depend on the model parameters (Table 5) collectively denoted as a vector $\boldsymbol{\Omega}$, and the bulk nucleotide concentrations c_T and c_D .

Model hyperparameters and parameters

In summary, our Markov model includes three hyperparameters. These hyperparameters have been introduced conceptually in Results, from which specific variants of our model are derived to test competing hypotheses, regarding:

- (1) The set of distinct β -subunit conformations (Three-conformation vs. Four-conformation). Does F₁-ATPase function require a minimal number of three functionally distinct conformations for individual β -subunits, or four conformations? The choices are $C^{(k)} \in \{o, h, c\}$ (*ohc*-variants), or $C^{(k)} \in \{o, h, c_1, c_2\}$ (*ohc*₁*c*₂-variants).
- (2) The catalytically active conformation(s) (One-active vs. Two-active). Within *ohc*₁*c*₂-variants, are both closed conformations (c_1, c_2) catalytically active, or only one? The choices are that either only c_1 is active (*ohc*₁*c*₂-variants), or both c_1 and c_2 are active (*ohc*₁*c*₂-variants).
- (3) The γ - β restrictions. How does the γ -subunit sterically restrict β -subunit conformations at each orientation? For *ohc*-variants, each $\Gamma_k^\phi = \{o\}, \{o, h\}$, or $\{o, h, c\}$; for *ohc*₁*c*₂-variants, each $\Gamma_k^\phi = \{o\}, \{o, h\}, \{o, h, c_1\}, \{o, h, c_2\}$, or $\{o, h, c_1, c_2\}$. Each combination of all six Γ_k^ϕ ultimately defines a specific variant (see, e.g., Tables 2 and 3).

Here, we introduce a more formal notation for the hyperparameter γ - β restrictions. Specifically, we define Γ_k^ϕ , the ensemble of conformations accessible to each β -subunit β_k at each γ -orientation ϕ . Due to the three-fold rotational pseudo-symmetry of F₁-ATPase structure, specifying six Γ_k^ϕ , e.g., $\{\Gamma_k^\phi | k = 1, 2, 3; \phi = 80^\circ, 120^\circ\}$, is sufficient to distinctly define a specific variant, as demonstrated in Fig. 2f and Tables 2, 3. Because these γ - β restrictions are defined to primarily model steric repulsion, a more open conformation should not be precluded when a more closed conformation is allowed. Accordingly, we assume that for *ohc*-variants, each Γ_k^ϕ can be $\{o\}$, $\{o, h\}$, or $\{o, h, c\}$. As a rough estimation, the combinations of the six Γ_k^ϕ give rise to $3^6 - 2^6 = 665$ *ohc*-variants; here, 2^6 combinations where c is completely missing are excluded. Similarly, for *ohc*₁*c*₂-variants, Γ_k^ϕ can be $\{o\}$, $\{o, h\}$, $\{o, h, c_1\}$, $\{o, h, c_2\}$, or $\{o, h, c_1, c_2\}$. Their combinations give rise to roughly $5^6 - 2^6 - (3^6 - 2^6) - (3^6 - 2^6) = 14,231$ *ohc*₁*c*₂-variants. Here, the subtractions exclude the 2^6 combinations where both c_1 and c_2 are completely missing, the $3^6 - 2^6$ combinations missing c_1 , and the $3^6 - 2^6$ combinations missing c_2 .

Finally, the parameters that are introduced into our model when defining the state free energies (Eqs. (2), (3)) and transition rates (Supplementary Table 2) are summarized in Table 5, including conformational free energies, binding free energies, and several free energy barriers. They count to 17 parameters for the *ohc*-variants and 22 parameters for the *ohc*₁*c*₂-variants (some are subjected to the constraint of Eq. (7)).

Derivation of statistical and kinetic observables from the non-equilibrium steady-state

Several statistical and kinetic observables are derived from the steady-state solution of the master equation (Eq. (2)), including (1) nucleotide occupancies, (2) populations of the 80° - and 120° -dwells, (3) turnover rates, rotation rates, and chemo-mechanical coupling efficiencies, and (4) the probabilities that a major catalytic step occurs at different γ -subunit orientations. These derived observables constitute the primary predictions of our model, and are systematically compared against experimental data during the training and cross-validation procedures (detailed in Results and Methods).

In this derivation, we assume the nucleotide concentrations c_T and c_D are constant. As mentioned in Results, this setup directly mimics typical experimental conditions (e.g., single-molecule experiments) where F₁-ATPase is in contact with external chemical reservoirs⁸. More generally, this assumption also serves as a valid approximation for typical bulk biochemical assays, where nucleotide concentrations are substantially higher than the enzyme concentration,^{19,21} rendering substrate depletion negligible. Consequently, this derivation describes an open system

driven into a NES) by the constant ATP/ADP chemical potential difference.

The steady-state populations of the Markov states under the NESS condition, $\mathbf{p}^{\text{st}}$, are obtained as the null space of the transition rate matrix $\mathbf{R}$, i.e., by solving the equation

$$\mathbf{R}(\Omega; c_T, c_D) \mathbf{p}^{\text{st}} = 0. \quad (9)$$

Although the steady-state populations $\mathbf{p}^{\text{st}}$ remain constant, the NESS condition established above allows a persistent net flux to flow through the network of states along thermodynamically favorable pathways. This net flux is defined for each transition between states i and j as:

$$f_{i \rightarrow j} = \rho_i^{\text{st}} r_{i \rightarrow j} - \rho_j^{\text{st}} r_{j \rightarrow i}. \quad (10)$$

It is the summation of these net fluxes over relevant transitions that generates the macroscopic observables like turnover and rotation rates, even though every elementary transition strictly satisfies local detailed balance (Eq. (6)).

Equation (9) is solved numerically by eigenvalue decomposition of the transition rate matrix $\mathbf{R}$. One eigenvalue of $\mathbf{R}$ is zero; the corresponding eigenvector, after normalization, gives the steady-state populations $\mathbf{p}^{\text{st}}$. Subsequently, $\{f_{i \rightarrow j}\}$ are calculated via Eq. (10). From these quantities, several basic observables can be directly evaluated, as summarized in Supplementary Table 6.

Based on these basic observables, we then define several composite observables that are used in the training and cross-validation procedures (Results). The total nucleotide occupancy ν is the sum of individual ATP and ADP occupancies:

$$\nu \equiv \nu_T + \nu_D = \sum_i (n_i^T + n_i^D) \rho_i^{\text{st}}. \quad (11)$$

The chemo-mechanical coupling efficiency η , a key measure of the motor's performance, is defined as the ratio of the rotational rate (scaled by 3 for a full cycle) to the turnover rate:

$$\eta = \frac{3k_{\text{rot}}}{k_{\text{cat}}}. \quad (12)$$

Finally, to compare our model prediction against the consensus chemo-mechanical coupling scheme (Table 6 and Supplementary Fig. 2) in the cross-validation procedure, we calculate the probability $P_{\phi}^{\Pi, k}$ that a major catalytic step Π (ATP binding, ATP hydrolysis, or product dissociation) occurs within the β_k -subunit at a given γ -orientation ϕ . This probability is calculated by normalizing the orientation- and site-specific rate $R_{\phi}^{\Pi, k}$ in Supplementary Table 6 by the average turnover rate per site ($k_{\text{cat}}/3$, considering the three-fold pseudo-symmetry of F_1 -ATPase):

$$P_{\phi}^{\Pi, k} = \frac{R_{\phi}^{\Pi, k}}{k_{\text{cat}}/3}. \quad (13)$$

The probabilities $\{P_{\phi}^{\Pi, k} | \phi = 80^\circ, 120^\circ, 200^\circ, 240^\circ, 320^\circ, 360^\circ\}$ visualized in Figs. 2e and 3a, b quantify how a catalytic step Π is distributed over all possible γ -orientations as catalysis proceeds in the given catalytic site, thus characterizing the stochastic nature of F_1 -ATPase rotary catalysis.

Systematic model comparison strategy

To objectively select the optimal model variant that is consistent with all experimental data on bacterial F_1 -ATPase, we employ a systematic model comparison strategy of training + cross-validation. This strategy

involves partitioning the available experimental data of bacterial F_1 -ATPase into two non-overlapping sets for training and cross-validation, respectively. These two sets of data, termed training data and validation data, are summarized in Table 6.

In the training procedure, we employ a Bayesian approach (detailed below) to broadly sample the high-dimensional parameter space (Table 5) and obtain a large ensemble of parameter sets that reproduce the training data within experimental uncertainty. These training data are quantitative measurements of several macroscopic properties that are fundamental to F_1 -ATPase function, namely turnover rates, chemo-mechanical coupling efficiency, and nucleotide titration curves. After training, an ensemble of parameter sets may be identified that reproduces the basic rotary catalysis behavior of F_1 -ATPase as core functional output.

Subsequently, we perform cross-validation by comparing the predictions of these trained parameter sets to the validation data. These validation data include qualitative to semi-quantitative observations that provide further insights into the mechanistic details of F_1 -ATPase on a more microscopic scale, mostly from single-molecule experiments, namely population shifts, consensus chemo-mechanical coupling scheme, and ratio between ADP and ATP occupancies. This cross-validation procedure allows us to further identify trained parameter sets that are also consistent with known mechanistic details. A model variant is considered successful only if it can reproduce both the training data and the validation data.

The rationale for this training + cross-validation strategy operates on two levels. Conceptually, they reflect a hierarchy in model evaluation. A minimal functional model must at least reproduce the training data, which are essential metrics that define F_1 -ATPase as a rotary molecular motor. Yet, there may be, and, as demonstrated in Results, there indeed are, multiple mechanistic scenarios on a microscopic level by which the similar rotary catalysis behavior on the macroscopic level can be achieved. Cross-validation is therefore performed to test which specific underlying mechanistic scenarios are consistent with the further experimental observations.

Practically, this strategy aligns with the quantitative nature of the data and the requirements of Bayesian parameter inference. The training data, being quantitative and often associated with well-defined experimental uncertainties, are readily incorporated into a statistically rigorous likelihood function (Eq. (15) and Supplementary Table 7), making them suitable for parameter inference. In contrast, for the qualitative or semi-quantitative validation data, formulating a precise likelihood term would be challenging and potentially introduce subjective assumptions.

We choose this training + cross-validation strategy over a formal, single-step Bayesian model comparison method (e.g., evaluating the marginal likelihood) for both pragmatic and scientific reasons. Pragmatically, an accurate estimation of the marginal likelihood requires extensive sampling of the high-dimensional posterior distribution, which is computationally infeasible for our model due to the high cost of evaluating the likelihood function for each parameter set.

More importantly, cross-validation is a scientifically necessary step to rigorously test a model's ability to generalize beyond simply fitting the data it is trained on and to validate its underlying physical assumptions. While Bayesian inference has a built-in mechanism to penalize model complexity (the Bayesian Occam's Razor), this mechanism operates under the assumption that the model framework is a reasonable representation of the data-generating process. If a model's foundational assumptions are flawed, a full Bayesian analysis might still yield a high-posterior model that has poor predictive power on unseen data—a form of model misspecification. Therefore, an external check like cross-validation remains a crucial step to ensure the robustness and physical realism of the model.

Table 6 | Experimental data used for training and cross-validation

Observable	Details	Source species
Training data		
Turnover rates	$k_{\text{cat}}([ATP])$ at $[ATP] = 1 \mu\text{M}$, $10 \mu\text{M}$, 0.1 mM , and 1 mM , derived from Michaelis-Menten constants $V_{\text{max}} = 80 \text{ s}^{-1}$ and $K_M = 0.04 \text{ mM}$ ¹⁹ .	<i>E. coli</i> *
Chemo-mechanical coupling efficiency	A chemo-mechanical coupling efficiency $\eta = 100\%$ is used as the training target for $[ATP]$ from micromolar to millimolar. This target value is consistent with the long-standing view of the enzyme as a near-perfect reversible motor ^{6,10} , and also strongly supported by the single-molecule data ^{8,54} .	Generally applicable
ATP/ADP titration curves	Measurements from tryptophan fluorescence experiments reported in Ref. 21 (data points recovered from Fig. 1A).	<i>E. coli</i> *
Validation data		
Population shift between 80°- and 120°-dwells	Single-molecule experiments show that the steady-state population of F_1 -ATPase shifts from the 120°-dwells (ATP-waiting dwells) to the 80°-dwells (catalytic dwells) as $[ATP]$ increases from sub-micromolar to millimolar levels ⁸ .	<i>Bacillus</i> PS3 [†]
Known γ -orientations of the major catalytic steps (consensus chemo-mechanical coupling scheme)	A consensus chemo-mechanical coupling scheme is synthesized from a broad range of experimental studies ^{8,15,20–22,32–34,54–57,65–67,70,75} (visualized in Supplementary Fig. 2; see also Supplementary Note 2), specifying the γ -orientations of the major catalytic steps: (i) under low $[ATP]$, ATP binding in a β -subunit occurs at a 120°-dwell (defined as 0° for reference); in this β -subunit, subsequently, (ii) ATP hydrolysis occurs at +200°; (iii) under high $[ATP]$, the rate-limiting product dissociation step occurs at +320°.	<i>E. coli</i> and <i>Bacillus</i> PS3
Ratio between ADP and ATP occupancies	Tryptophan fluorescence experiments show that at saturating $[ATP]$, the ratio of bound ADP to bound ATP ($(\nu_D : \nu_T)$) is ~2:1 ²⁰ .	<i>E. coli</i> [§]

All experimental data used for training and cross-validation were obtained from studies on F_1 -ATPase from two bacterial species, *Escherichia coli* (*E. coli*) and the thermophilic *Bacillus* PS3 (EF₁, TF₁). Both EF₁ and TF₁ have been shown to exhibit a similar well-characterised 80°/40° sub-stepping pattern^{8,15,59}, suggesting mechanistic consistency. Note that some studies employed specific F_1 -ATPase mutants (detailed in the footnotes below); however, these mutations have been shown to have only minor effects on enzymatic activity^{8,13,14,19,20}.

*Data obtained using *E. coli* F_1 -ATPase carrying the β -Y331W mutations^{19,21}.

†Data obtained using *Bacillus* PS3 F_1 -ATPase carrying the following mutations: γ -S107C, γ -I210C, and α -C193S; an extra his-tag (consisting of 10 histidines) at the amino terminus of each β -subunit^{8,13,14}.

§Data obtained using *E. coli* F_1 -ATPase carrying the β -F148W mutations²⁰.

Bayesian training approach

The aim of our Bayesian training approach is to sample an ensemble of parameter sets (Table 5) that adequately reproduce the training data (Table 6). According to the Bayes theorem, the posterior probability, $P(\Omega|\Psi^{\text{exp}})$, i.e., the probability of a parameter set Ω given the training data Ψ^{exp} , is proportional to the likelihood $P(\Psi^{\text{exp}}|\Omega)$ and the prior probability $P(\Omega)$:

$$P(\Omega|\Psi^{\text{exp}}) \propto P(\Psi^{\text{exp}}|\Omega) \cdot P(\Omega). \quad (14)$$

Here, the likelihood quantifies probability of the training data given the parameters, while the prior defines our initial assumptions about the parameters. Assuming statistical independence of all data points in Ψ^{exp} and of all parameters, we define the likelihood function $P(\Psi^{\text{exp}}|\Omega)$ as the product of contributions from the three datasets (turnover rates $\{k_{\text{cat},i}^{\text{exp}}\}$, chemo-mechanical coupling efficiencies $\{\eta_i^{\text{exp}}\}$, and nucleotide occupancies $\{\nu_i^{\text{exp}}\}$), assuming statistical independence of all data points:

$$P(\Psi^{\text{exp}}|\Omega) = \prod_l P(k_{\text{cat},l}^{\text{exp}}|\Omega) \cdot \prod_l P(\eta_l^{\text{exp}}|\Omega) \cdot \prod_l P(\nu_l^{\text{exp}}|\Omega), \quad (15)$$

where the product index l runs over all data points within each respective dataset; the numbers of data points in the three datasets may differ. Supplementary Table 7 details the specific distribution assumed for each component. Similarly, we define the prior distribution $P(\Omega)$ as the product of the prior distributions of each individual parameter:

$$P(\Omega) = \prod_{n=1}^N P(\omega_n). \quad (16)$$

We assume a uniform distribution for each individual prior distributions $P(\omega_n)$ (detailed in Supplementary Table 1). This choice is equivalent to assuming log-uniform priors for the transition rates that

are determined by these parameters (free energies and free energy barriers) via the Boltzmann factor, ensuring that no a priori bias towards any particular timescale is imposed for a transition rate. The ranges for our uniform priors of the parameters (Supplementary Table 1) were chosen to restrain the resulting rates to biophysically plausible limits, e.g., by precluding conformational transitions faster than the nanosecond timescale.

Critically, our goal is not to identify a single maximum a posteriori (MAP) estimate of the parameters. Instead, acknowledging the potential complexity of the posterior landscape, which might contain multiple local maxima, we aim to sample an ensemble of high-posterior parameter sets that broadly covers these distinct regions. To this end, we perform hundreds of independent optimization runs, each initiated from a different, randomly chosen starting point. A stochastic hill-climbing search algorithm is employed for each individual optimization run, where the parameter set Ω stochastically explores the parameter space, each step increasing the posterior, until a maximum is reached. At each step within an optimization run, denoting the current parameter set as Ω^{now} , a number N_{try} of guesses $\{\Omega^{\text{try}}\}$ are sampled, where each individual parameter ω_i^{try} is drawn from a Gaussian distribution centered at ω_i^{now} with standard deviation σ_{try} . When some Ω^{try} are found to give higher posteriors than Ω^{now} , the parameter set Ω then moves to the Ω^{try} of the highest posterior. Otherwise, the parameter set Ω stays at Ω^{now} , and σ_{try} is tuned down by a factor ϵ ($0 < \epsilon < 1$) to allow searching within a smaller area around Ω^{now} , which is likely near a local maximum.

Finally, all resulting parameter sets that adequately reproduce the training data are collected into an ensemble (trained parameter sets) and carried forward for the subsequent cross-validation procedure.

Evaluation of nucleotide titration curves and derivation of apparent binding affinities

Here we detail the calculation and analysis of nucleotide titration curves, which are crucial for evaluating our model against

experimental data (Results). The nucleotide titration curve represents the steady-state total nucleotide occupancy ν (Eq. (11)) as a function of nucleotide concentration. Mimicking experimental protocols for ATP titration curves (where [ADP] is kept low while [ATP] is varied from nanomolar to millimolar ranges)²¹, we calculated the ATP titration curve numerically as part of the training procedure (Results). This calculation involved repeatedly solving for the steady-state distribution $\mathbf{p}^{\text{st}}$ (Eq. (9)) while varying [ATP] from 100 nM to 1 mM (assuming a constant [ADP] of 1 nM), and then evaluating the total occupancy ν (Eq. (11)). Similarly, an ADP titration curve was evaluated by varying [ADP] over the same range (assuming a constant [ATP] of 1 nM).

To quantitatively assess the agreement between our model-predicted ADP titration curves and the experimentally measured one²¹, we utilize the apparent ADP binding affinities derived from fitting both curves to an equilibrium binding model^{20–22}:

$$\nu([\text{ADP}]) = \frac{[\text{ADP}]}{[\text{ADP}] + K_{d1}} + \frac{[\text{ADP}]}{[\text{ADP}] + K_{d2}} + \frac{[\text{ADP}]}{[\text{ADP}] + K_{d3}}. \quad (17)$$

The three fitting parameters, $K_{d1,2,3}$, resulting from this analysis are interpreted as (apparent) ADP binding affinities (dissociation constants) of the three β -subunits. They quantitatively characterize the shape of the titration curve, as their relative magnitudes determine the number of distinct binding steps identifiable therein. The experimentally determined values exhibit distinct magnitudes ($K_{d1} = 41$ nM, $K_{d2} = 6.0$ μ M, $K_{d3} = 42$ μ M)²¹, providing the specific benchmark against which our model predictions are compared (Results).

Notably, these apparent binding affinities $K_{d1,2,3}$ obtained from fitting (Eq. (17)) are emergent, macroscopic properties reflecting the ensemble behaviors of the three differentiated β -subunits. Given the physical picture underlying our model where each β -subunit dynamically transitions between multiple conformations, an apparent affinity K_{dk} does not necessarily correspond to the intrinsic thermodynamic property of a single, static conformation. The latter, which we term a microscopic binding affinity, is instead directly related to a specific conformation C and defined via the standard thermodynamic relation

$$K_{d,c} \equiv \exp\left(\frac{\Delta G_{b,c,D}^{\circ}}{k_B T}\right), \quad (18)$$

where $\Delta G_{b,c,D}^{\circ}$ represents the standard ADP binding free energy of conformation C , one of our model parameters (Table 5).

We establish a theoretical link between the apparent affinities $K_{d1,2,3}$ and the microscopic affinities $\{K_{d,c}\}$, via an analytical expression derived in Supplementary Note 5. This analytical expression provides the theoretical foundation for our understanding of the incompatibility of the *ohc-w* and *ohc-m* variants with the experimentally determined apparent ADP binding affinities (Results). Below, we outline the key steps of this derivation.

First, we derive an approximate analytical solution for the steady-state distribution $\mathbf{p}^{\text{st}}$ (Supplementary Equations (48)–(53)). Under the low [ATP], when evaluating an ADP titration curve, the turnover rate is approximately zero, and the system approaches equilibrium (equilibrium approximation). Thus, the steady-state distribution approximates a Boltzmann distribution, which can be analytically expressed in terms of the underlying free energies of the respective Markov states (Eq. (2)). These state free energies, in turn, depend on our model parameters, our model parameters (specifically, the conformational free energies $\{\Delta G_{\beta,c}\}$ and the standard ADP binding free energies $\{\Delta G_{b,c,D}^{\circ}\}$, Table 5). Using this analytical expression for the steady-state distribution, the ADP titration curve $\nu([\text{ADP}])$, and subsequently the

apparent ADP binding affinities derived from it, can also be analytically expressed in terms of these model parameters (Supplementary Equations (55)–(59)).

Particularly, for model variants where the three β -subunits are decoupled (i.e., $\Gamma_k^{80^\circ} = \Gamma_k^{120^\circ}$ for at least two β -subunits, see Supplementary Note 5 for explanation), e.g., variants *ohc-w/s*, the analytical expressions for the apparent ADP binding affinities are rather straightforward. Each apparent ADP binding affinity K_{dk} is expressed as a superposition of the microscopic binding affinities

$K_{d,c} = \exp\left(\frac{\Delta G_{b,c,D}^{\circ}}{k_B T}\right)$ of the several accessible conformations,

$$(K_{dk})^{-1} = \sum_{(\phi,C)} w_{(\phi,C)} (K_{d,c})^{-1}, \quad (19)$$

where the summation is over all combinations of ϕ and $C \in \Gamma_k^{\phi}$; the weights $\{w_{(\phi,C)}\}$ depend on the conformational free energies $\{G_{\beta,c}\}$ of the accessible conformations (see Supplementary Equations (77) and (78)).

Physical constraints on γ - β restrictions

To systematically reduce the vast number of potential model variants (~ 600 for *ohc*-variants and $\sim 14,000$ for *ohc*₁*c*₂-variants), we derive several necessary conditions of γ - β restrictions for our model to be compatible with the consensus chemo-mechanical coupling scheme (Table 6 and Supplementary Fig. 2), as summarized in Supplementary Table 4 and detailed in Supplementary Note 3. These conditions are the technical implementation of the fundamental binding-change principle^{6,15–17,35,68,73}, complemented by our finding that a nucleotide-bound β -subunit energetically favors the catalytically active *c* conformation (both discussed in Results). For example, for ATP hydrolysis to occur in a β -subunit at $+200^\circ$ after ATP binding, the γ - β restrictions must allow this β -subunit to adopt the catalytically active *c* conformation at $+200^\circ$, but then enforce its transition into an inactive conformation (*h* or *o*) upon rotation to $+240^\circ$ (Supplementary Equations (8) and (9)). If *c* remained accessible at $+240^\circ$, this energetically favored closed state would likely persist, potentially stalling the cycle or preventing subsequent steps, e.g., product dissociation preparation, which require fewer closed conformations. Applying this allow-and-enforce logic to all major catalytic steps yields the set of conditions summarized in Supplementary Table 4.

Note that all Family B variants (Supplementary Equation (7) in Supplementary Table 4) are further excluded as reasoned in Results. To further reduce the choices of plausible *ohc*₁*c*₂-variants, we introduce several additional constraints, considering the resolved structures of F₁-ATPase, the physical parsimony of our aimed minimal model, and the intrinsic structural asymmetry of the γ -subunit (Supplementary Table 5). More detailed explanations for these constraints are provided in Supplementary Notes 3&4.

KMC simulations of F₁-ATPase coupled to external load

KMC simulations mimicking the single-molecule experiments on F₁-ATPase are presented in Results (see particularly Fig. 4a). In these simulations, the rotation of the γ -subunit and the circular motion of the probe centroid are coupled by a harmonic potential with force constant κ :

$$V(\phi, \phi_p) = \frac{1}{2} \kappa (\phi - \phi_p)^2, \quad (20)$$

where ϕ and ϕ_p are the angular positions of the γ -subunit and the probe centroid, respectively. Further, the probe is subjected to the viscous friction of the solution with a rotational friction drag

coefficient ξ . We estimate the spring force constant $\kappa \approx 5 k_B T$ and the rotational friction drag coefficient $\xi \approx 6.7 \times 10^{-5} k_B T \cdot s$ for a 20-nm colloidal gold bead⁸. κ is estimated from the widths of the distributions of the recorded probe positions. ξ corresponds to the midpoint of $8\pi\mu r^3 \leq \xi \leq 14\pi\mu r^3$ estimated in ref. 8 (bead radius $r = 20$ nm, water viscosity $\mu = 10^{-9}$ pN · nm⁻² · s).

Given the estimated values of κ and ξ , the inertia of the probe is negligible compared to the viscous friction and force from the spring. Therefore, we use the OLE to describe the circular motions of the probe:

$$\xi \frac{d\phi_p}{dt} = -\frac{\partial V}{\partial \phi_p} + \tau_B(t), \quad (21)$$

where $\tau_B(t)$ is a fluctuating Brownian torque. This torque has the statistical properties $\langle \tau_B(t) \rangle = 0$ and $\langle \tau_B(t) \tau_B(t') \rangle = 2\xi k_B T \delta(t - t')$, where $\delta(t - t')$ is the Dirac delta function.

We simulate the coupled system by combining stochastic integration of the OLE (Eq. (21)) with Gillespie algorithm^{85,86} for our Markov model of F₁-ATPase. Consider the n th step starting at time t_n , when the probe is at $\phi_p(t_n)$, and F₁-ATPase is in Markov state $s_n = i$, with γ -orientation ϕ_n . We assume that F₁-ATPase remains in this state s_n during this step, which lasts until time t_{n+1} when the next step initiates, and F₁-ATPase jumps to another state s_{n+1} instantaneously. During this time interval (t_n to t_{n+1}), the γ -subunit remains at ϕ_n , while the probe moves according to Eq. (21). The probe position at the end of the interval, $\phi_p(t_{n+1})$, is calculated via stochastic integration using the Euler-Maruyama method:

$$\phi_p(t + \Delta t) = \phi_p(t) - \frac{\kappa}{\xi} (\phi_p(t) - \phi_n) \Delta t + \epsilon \sqrt{\frac{2k_B T}{\xi}} \Delta t. \quad (22)$$

Here, ϵ is a random number drawn from the standard normal distribution. This equation is iteratively evaluated with a small time step Δt from $t = t_n$ until t_{n+1} . We use $\Delta t = 10^{-9}$ s, which is at least one magnitude smaller than the timescale of the fastest transition in our Markov model.

The next state s_{n+1} of F₁-ATPase and the holding time $\tau = t_{n+1} - t_n$ are determined by Gillespie algorithm^{85,86}. The next state s_{n+1} is chosen from the set of states $\{j\}$ accessible from the current state i , with probability $P(s_{n+1} = j) = r'_{i \rightarrow j} / r'_i$, where $r'_{i \rightarrow j}$ is the transition rate from i to j for F₁-ATPase under load, and $r'_i = -\sum_j r'_{i \rightarrow j}$ is the total rate of leaving state i . τ is sampled from the exponential distribution $\tau \sim r'_i \exp(-r'_i \tau)$. Crucially, these transition rates $\{r'_{i \rightarrow j}\}$ differ from the rates $\{r_{i \rightarrow j}\}$ defined earlier (Supplementary Equations (1)–(5)) for unloaded F₁-ATPase, because the harmonic potential coupling the γ -subunit and the probe modifies the free energy landscape. For transitions where the γ -subunit does not rotate, the landscape is unaffected, and thus $r'_{i \rightarrow j} = r_{i \rightarrow j}$.

In contrast, for a transition involving γ -subunit rotation by $\Delta\phi$, the harmonic potential $V(\phi, \phi_p)$ alters the free energy profile along the rotation coordinate ϕ . Taking the initial state i (at ϕ_n) as the reference, the free energy profile becomes

$$\Delta G'_\gamma(\phi) = \Delta G_\gamma(\phi) + V(\phi, \phi_p) - V(\phi_n, \phi_p), \quad (23)$$

where $\Delta G_\gamma(\phi)$ is the profile for unloaded F₁-ATPase. Assuming the transition state is located symmetrically between the initial and final states, i.e., at $\phi^* = \phi_n + \lambda \Delta\phi$ with $\lambda = 0.5$, the activation energy barrier $\Delta G^*_{\gamma'} (\equiv \Delta G'_\gamma(\phi^*))$ relative to the initial state is modified from the unloaded barrier ΔG^*_γ according to

$$\Delta G^*_{\gamma'} = \Delta G^*_\gamma + \frac{1}{2} \kappa ((\phi_n + \lambda \Delta\phi - \phi_p)^2 - (\phi_n - \phi_p)^2). \quad (24)$$

Consequently, the transition rate under load becomes:

$$r'_{i \rightarrow j} = r_{i \rightarrow j} \exp\left(-\frac{\kappa}{2k_B T} ((\phi_n + \lambda \Delta\phi - \phi_p)^2 - (\phi_n - \phi_p)^2)\right). \quad (25)$$

Hidden Markov analysis of the simulated single-molecule trajectories

To extract the dwell positions and kinetics (Fig. 4e–g) from the simulated KMC trajectories, we employ hidden Markov analysis (HMA)⁸⁷. The detailed mathematical formulation is provided in Supplementary Note 6, while an overview is presented below.

The goal of HMA is to infer the most likely sequence of underlying discrete dwell states from the simulated trajectory of the probe's angular position ϕ_p , and subsequently to obtain the lifetime distributions for each dwell. Our HMA assumes the system transitions between a small number of discrete dwell states following Markovian dynamics, where transitions are restricted to adjacent dwell states (Supplementary Equations (90) and (91)). The observed probe position ϕ_p at any time is assumed to follow a von Mises distribution centered around the true dwell angle (Supplementary Equation (92)).

We first use a modified Baum-Welch algorithm, adapted for the three-fold rotational symmetry of F₁-ATPase (Supplementary Equations (94)–(102)), to estimate the HMA parameters, including the true dwell angles, the angular variances, and the transition probabilities. After obtaining the maximum-likelihood parameters, we use the Viterbi algorithm to infer the most probable sequence of dwell states from the simulated trajectory (Supplementary Equations (103)–(104)). Based on this inferred sequence, the trajectory is segmented to generate the dwell lifetime distributions shown in Fig. 4e–g. These distributions are then fitted by a single exponential function $f(t) \propto \exp(-rt)$ or a double exponential function $f(t) \propto \exp(-r_1 t) - \exp(-r_2 t)$ to determine the number of rate-limiting steps and their rate constants.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The previously published atomic coordinates of F₁-ATPase referred to in this study are available in the Protein Data Bank (PDB) under accession code **1BMF**. Source data are provided with this paper.

Code availability

The custom code developed in this study is publicly available on the Github repository YixinChen95/MarkovianF1 (<https://github.com/YixinChen95/MarkovianF1>), and archived at Zenodo (<https://doi.org/10.5281/zenodo.19133448>)⁸⁸. The repository contains the C and Python scripts organized into four modules for Bayesian training of the Markov model, evaluation of model predictions, kinetic Monte-Carlo simulations, and HMA of the simulated trajectories, respectively. A README file provides instructions for installation and execution, and includes example workflows demonstrating the main functions of the code.

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Author contributions

H.G. conceived the project. Y.C. conducted the research. Y.C. and H.G. interpreted the results and wrote the paper.

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Competing interests

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Additional information

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